LONDON METROPOLITAN UNIVERSITY

islington college
(इस्लिङ्टन कलेज)

Module Code & Module Title

CS6P05NI Project

Interim Report

25% Individual Coursework

Submission: Final Submission

AY 2025 2026

Academic Semester: Autumn Semester 2025

Credit: 30 Credit Year Long Module

Student Name: Puspanjali Adhikari

London Met ID: 23049017

College ID: np01cp4a230460

Assignment Due Date: Tuesday, December 30, 2025

Assignment Submission Date: Tuesday, December 30, 2025

Internal Supervisor: Mr. Bikram Poudel

External Supervisor: Mr. Mohit Paudel

I confirm that I understand my coursework needs to be submitted online via MST Classroom under the relevant module page before the deadline for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.

Summary

This interim report outlines the developments and results of the AidFlow project, that is an established donation management system and aimed at bringing transparency, accountability, and trust to the Nepalese charity ecosystem. Some of the major milestones are the accomplishment of the requirement analysis, system architecture, UML diagrams, ERD, flowcharts, DFDs, wireframes, and the creation of the MERN-based develop environment. The project is currently underway as the backend structure and database schemas have already been initiated in accordance with the Agile Scrum schedule. The second stage will be aimed at core module implementation, frontend and backend component integration, and major workflow testing. After reaching maturity, AidFlow will be a single and trusted platform that will increase the donor's confidence and help the NGOs with the verified, transparent and traceable project management.

Table of Contents

1. Introduction 1
1.1 Current scenario 2
1.2 Problem Statement 2
1.3 Project as a solution 2
1.4 Project Aim and objectives 3
1.4.1 Aim: 3
1.4.2 Objectives 3
2. Background and Literature Review 5
2.1 About the End User/Client 5
2.2 Understanding the Project 5
2.3 Similar Project 6
2.3.1 GoFundMe 6
2.3.2 Global Giving 6
2.3.3 Just Giving 7
2.3.4 Milaap 8
2.4 Comparison Analysis of Similar Projects 9
3.Development Methodology 10
3.1 Considered methodology 10
3.2 Used Methodology 10
3.3 Gantt Chart 11
3.4 Requirement Gathering 13
4. Development to Date 14
4.1 System Architecture Design 14
4.2 Use Case Diagram 15
4.3 Data Flow Diagram 15
4.4 Class diagram 17
4.5 Entity Relationship Diagram 18
4.6 Sequence Diagram 19
4.6.1 Sequence diagram of Login 19
4.6.2 Sequence diagram of NGO Registration 20

4.6.3 Sequence diagram of Donation Process 21
4.6.4 Sequence diagram of Project Update 22
4.7. Flow chart 23
4.7.1 Flowchart of Login 23
4.7.2 Flowchart of NGO registration 24
4.7.3 Flowchart of Donation Process 25
4.7.4 Flowchart of Project Update 26
4.8 Wire Frame 27
4.9 UI Design 29
4.10 Development status with Evidence 38
5.Analysis of progress 40
5.1 Progress Review 40
5.2 Progress Table 40
5.3 Action Plan 42
6 Future Work 43
6.1 Remaining Task Analysis with Gantt chart 43
6.1.1 For Further Future Work 43
Conclusion 44
Bibliography 45
Appendix 46
8.1 Appendix 1: System/Software Requirements (SRS) 46
8.1.1 Introduction 46
8.1.1.1 Purpose 46
8.1.1.2 Project Scope 46
8.1.2 Overall Description 47
8.1.2.1 System Perspective Diagram 47
8.1.2.2 System Features 47
8.1.3 Functional Requirements 50
8.1.4 External Interfaces 51
8.1.4.1 User Interface 51
8.1.4.2 Hardware Interfaces 51
8.1.4.3 Software Interfaces 52

8.1.4.4 Communication Interfaces 52
8.1.4.5 Non-Functional requirements 52
8.1.4.6 Safety Requirements 52
8.3 Appendix 3: High Level and Expanded Level Descriptions 53
8.4 Appendix 4: Future Work 56
8.4.1 Remaining Phases 56
8.4.2 Detailed Plan 57
8.5 Appendix 5: Survey Form 57
8.6 Appendix 6: Used Methodology 60
8.7 Appendix 7: Problem Statement 61

Table of Figure

Figure 1 GoFundMe website 6
Figure 2 GlobalGiving Website 7
Figure 3 JustGiving Website 7
Figure 4 Milaap Website 8
Figure 5 WBS 11
Figure 6 Gantt Chart 11
Figure 7 Gantt Chart 12
Figure 8 System Architecture Design 14
Figure 9 Use Case Diagram 15
Figure 10 Data Flow Diagram level 0 15
Figure 11 Data Flow Diagram Level 1 16
Figure 12 Class diagram 17
Figure 13 Entity Relationship Diagram 18
Figure 14 Sequence diagram of Login 19
Figure 15 Sequence diagram of NGO Registration 20
Figure 16 Sequence diagram of Donation Process 21
Figure 17 Sequence diagram of Project Update 22
Figure 18 Flowchart of Login 23
Figure 19 Flowchart of NGO registration 24
Figure 20 Flowchart of Donation Process 25
Figure 21 Flowchart of Project Update 26
Figure 22 Wireframe home page 27
Figure 23 Donate Portal 27
Figure 24 Enhanced Donation Management 28
Figure 25 NGO Verification 28
Figure 26 UI Login 29
Figure 27 UI Login 29
Figure 28 AidFlow Home page 30
Figure 29 AidFlow Browse page 30
Figure 30 AidFlow NGO page 31

Figure 31 NGO Dashboard 32
Figure 32 Admin Dashboard 32
Figure 33 Aid flow project page 33
Figure 34 NGO Dashboard 34
Figure 35 Admin Dashboard 35
Figure 36 badge Showcase 35
Figure 37 Map Integration 36
Figure 38 NGO Home Page 37
Figure 39 Login Code 38
Figure 40 Login Code 38
Figure 41 Login Code 39
Figure 42 Login Code 39
Figure 43 Remaining Work Gantt chart 43
Figure 44 Overall Description 47
Figure 45 Survey Form 57
Figure 46 Survey Form 58
Figure 47 Survey Form 58
Figure 48 Survey Form 59
Figure 49 Survey Form 59
Figure 50 Survey Form 60

Table of Table

Table 1 Comparison table 8
Table 2 Progress Table 40
Table 3 Registration & Login 50
Table 4 Project and Donation 50
Table 5 Verification System 51
Table 6 Transparency & Impact Reporting 51
Table 7 Expanded Use case 55
Table 8 Expanded Use case 55

1. Introduction

In Nepal where philanthropy is already a well established culture due to the high Sewa, or tradition of charity, and community support in the country, philanthropic activities are gaining popularity. The recent rise of local and diaspora charity is due to the recent natural disasters such as the earthquake in 2015 and the COVID-19 pandemic. Nevertheless, the delivery system of donations is still unbelievably fragmented and full of suspicion in Nepal. The problem of transparency deficiency, the inability to check the credibility of various local NGOs and institutions, and the absence of a reliable system that could oversee the contributions of the good-intentioned individuals are discussed (Bhandari, 2022).

Donors often find it hard to find decent organizations, schools or local projects which require support. Most people who give charity are left to wonder what becomes of their donations as there are no updates or scanty updates to the progress of the project or the use of the funds given. It is noncompliance, which discourages further participation due to the atmosphere of uncertainty and distrust. (Prakash, 30 March 2010).

One of the problems that will be trying to be addressed by this project is the establishment of a unified transparent donation management system. The system will enable funders to monitor their money in real time and get a lot of information concerning their needs matched with reputable NGOs. The technological solution will have a well-formatted platform that will enhance the legitimacy, effectiveness, and accessibility of charitable giving among people and organisations that intend to make major charitable contributions. (Charities Aid Foundation, 2023).

Addressing the poor areas that render the ecology of the Nepalese donation so fragmented, the project is aimed to recover the confidence, the transparency, and the more consistent flow of the charitable donations to the country.

1.1 Current scenario

It is difficult to assure the donors that the philanthropical giving in Nepal is not scattered, disorganised and opaque. Most individuals are willing to give to local non-governmental organisations, schools and charities, yet they often fail to locate reputable organisations, where to ensure that their money is actually spent, or to discover the outcome of their donation. This information is lacking and this hinders any long-term or frequent donations, and this is one of the greatest impediments to trust. (Shrestha, 2019).

Most NGOs in Nepal continue to use word-of-mouth, social media updates and manual reporting; these forms do not provide a coherent, verifiable and consistent image of the donor action. This means that the number of individuals involved in sustainable donations is reduced. (Bhandari, 2022).

1.2 Problem Statement

Principal Obstacles

- Donors can hardly find trustworthy and properly researched NGOs or educational institutions.
- The openness in terms of spending the donations is not very effective.
- The donors will not receive regular post-gift comments or progress, or updates on the achievements of their donation.
- This will result in ineffective operations and false information as there will be no single open ecosystem between donors, NGOs and community activities.

For Supporting Data

1.3 Project as a solution

This site unites donation under a single platform through provision of:

- The integrated platform of the system equally offers easy access to reliable, accredited non-governmental organisations, and community organisations by the donors.

- The software is also open minded in the use of the donations because it manages the entire donation process by adding the receipts, evidence, time and place information.

- The donors will not stay in darkness after donating since they will be availed with the logs of the activities, pictures and real time updates of the progress.

- In clear requirements and updated programs, it is only under such circumstances that donors can be capable of obtaining reliable, systematic and valid information at the NGOs before they can make their donations.

- This is attributed to the fact that stringent vetting process will be carried out to ensure that it will only allow real and credible NGOs and institutions to appear on the platform.

- The financial and operational activities are easily monitored with the help of transparent reports of use including bills, before and after photographs, completion signs.

This enhances confidence, notion of access and transparency to the whole system, providing of charities becomes more reliable, sizable and accountable.

1.4 Project Aim and objectives

1.4.1 Aim:

To create a web and mobile platform that will result in centralisation and increase the participation of the donor in the process of donating to charities in Nepal through increased efficiency and transparency in the process.

1.4.2 Objectives

Specific: To create and roll out a safe donor portal whereby there are options of a one time and a recurring Donation, becoming a user, and having a profile.

Achievable: To develop a special platform of the NGOs, which will assist them enlist their requirements, construct authenticated profiles, and will present at least two updates on the development of every funded project.

Realistic: Include an interactive map that will allow the contributors to search and filter at least thirty certified groups based on their location and cause with the help of the Google Maps API.

Time bound: The prototype with a gamification system (badges, points, and a scoreboard) would have to be developed by the end of the development process and be tested in relation to the user engagement.

2. Background and Literature Review

2.1 About the End User/Client

Charitable giving in other countries is especially important to developing countries with limited resources, where it enhances social welfare, healthcare, education, and community development. Nonetheless, donors may not be able to locate credible NGOs, trace the way their funds are being utilized, or they may not receive relevant information once they have donated. Those gaps make the donor to wonder if their fund is going to the needy, they intended to give when many of the NGOs are doing meaningful work. Due to the lack of an organized digital structure, it is hard to keep accountable or transparent as communication is spread among emails, social media posts, and informal updates.

2.2 Understanding the Project

Creating a transparent, structured, and trusted way of connecting donors with authentic NGOs, AidFlow is filling a significant gap in the donation system. The traditional modes of donation are often not transparent, and this is why the donors often struggle to settle on credible organizations, track how their funds are used, or determine the actual effect of their help. Likewise, NGOs find it difficult to be visible, retain the confidence of the donors as well as present their work in a modern digital form. AidFlow provides an improved technologically superior approach, which enhances accountability, transparency, and communication between all stakeholders. To offer the efficient and reliable donation experience, AidFlow includes such characteristics as project tags, project visual dashboards, impact reports, and secure online payment.

The Key Features of AidFlow are the following:

- AidFlow serves as a major point of centrality where donors can visit NGOs, explore the current projects, and donate through secure online systems.
- Donors are informed in advance about the specifics of the project including the objectives, budgets, updates, and verification of a project before making donation.
- To maintain accountability and donor trust, NGOs have an opportunity to post documents, project updates or communicate progress.

- The system enables the donors to monitor the developments of project in real time.
- AidFlow has a check system that ensures that it only publishes real project and NGOs.

2.3 Similar Project

The given below projects tackle the similar issues. By looking into those projects, I was able to learn more about how to improve my own product and make it work well for the user.

2.3.1 GoFundMe

GoFundMe is considered one of the largest crowdfunding platforms in the world that help individuals and organization to raise funds towards community projects, emergencies, education, health, and personal needs (Gofundme, 2010-2025).

Figure 1 GoFundMe website

2.3.2 Global Giving

GlobalGiving is an international donor network that connects donors to well-known nonprofits all over the world. It offers impact reports, organisation biographies and project listing (Giving, n.d.).

Figure 2 GlobalGiving Website

2.3.3 Just Giving

JustGiving, which is an international online fundraising platform, is used by individuals, charities, and nonprofits to raise funds to support humanitarian, social and personal interests. It allows confirmed organizations to establish project locations, recruit donors, and update followers on progress (JustGiving, 2017).

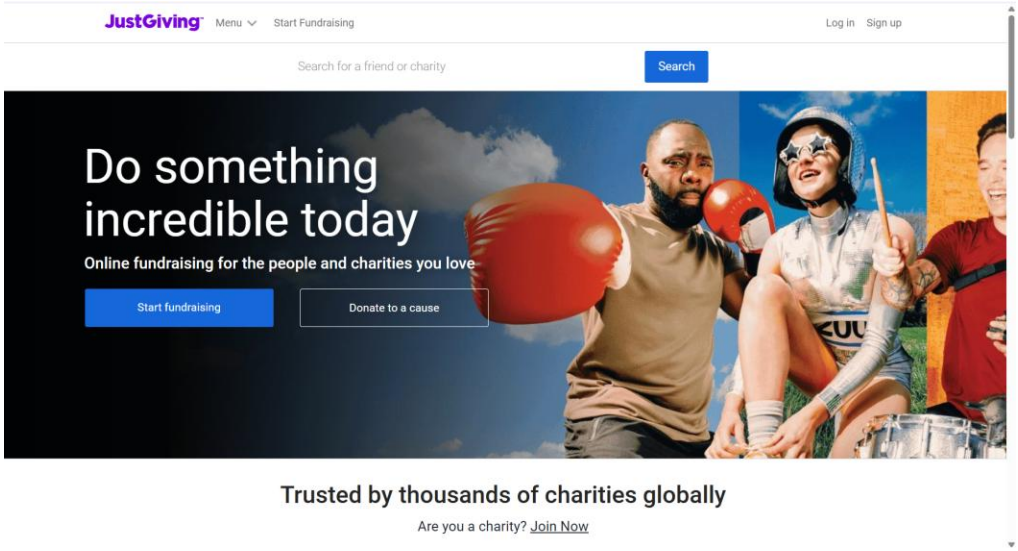

Figure 3 JustGiving Website

2.3.4 Milaap

The largest crowdfunding platform of personal and community causes in South Asia is called Milaap, and its causes include healthcare, education, women empowerment, and development of local community. It enjoys a huge following in India and Nepal and therefore a good comparison to AidFlow (Milaap, 2010).

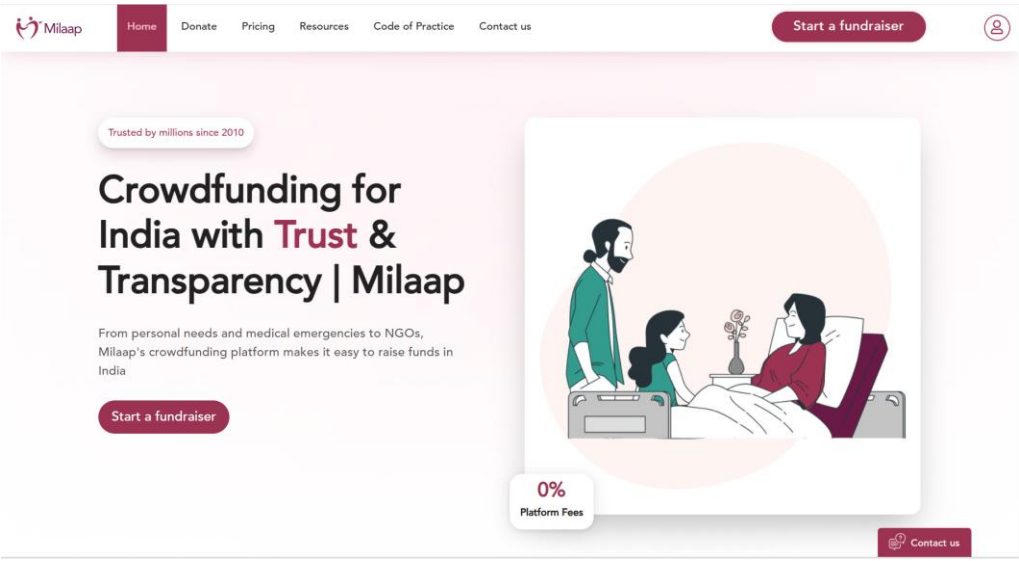

Figure 4 Milaap Website

Table 1 Comparison table

S. N	Feature	AidFlow	GoFundMe	GlobalGiving	JustGiving	Milaap
1	NGO Verification	✓	✗	✓	✗	✗
2	Real time project updates	✓	✗	✓	✗	✗
3	Transparent donation tracking	✓	✗	✓	✗	✗

4	Secure digital payment	✓	✓	✓	✓	✓
5	Impactful Dashboard	✓	✗	✓	✗	✗
6	Nepal Focused Support	✓	✗	✗	✗	✗

2.4 Comparison Analysis of Similar Projects

From the comparison we can clearly see that most of this fund raiser project do have few features, but they do not provide the transparency. And they do not provide a combined solution that includes verification, real time updates, transparency, secure digital payment, impactful dashboard, Nepal focused support. This is where The AidFlow stands to fill all this desire gaps in between.

3.Development Methodology

The Donation Management System (DMS) was to be planned, designed, developed, and tested in a systematic approach to Software Development Life Cycle. SDLC provides a very rigid guideline that helps to carry out each phase of the project in a systematic manner to help in requirement analysis as well as iterative development. (Sommerville, 2016).

3.1 Considered methodology

Three Different methodologies were considered before finalizing one for AidFlow .

a. **Prototype Methodology**

The prototype approach was initially considered because of the possibility to work out the initial versions of the system elements and obtain the feedback of the users before the full development process (Sommerville, 2016). The NGOs and donors may require prototypes in visualising dashboards, listing and verification of workflows in the DMS case. Such an approach was not however picked since it was time consuming.

b. **Spiral methodology**

Spiral approach was also a considered approach as it is based on risk analysis, a looped development, and constant improvement. It can be applied in complex features like secure payment integration and gift monitoring with its systematic cycles of planning, prototyping and assessment.

However, the large number of reviews processes and the lengthy documentation needed by the Spiral model can be time consuming to a single developer with little time to spare.

3.2 Used Methodology

For Description Click here Agile Methodology (Scrum Framework)

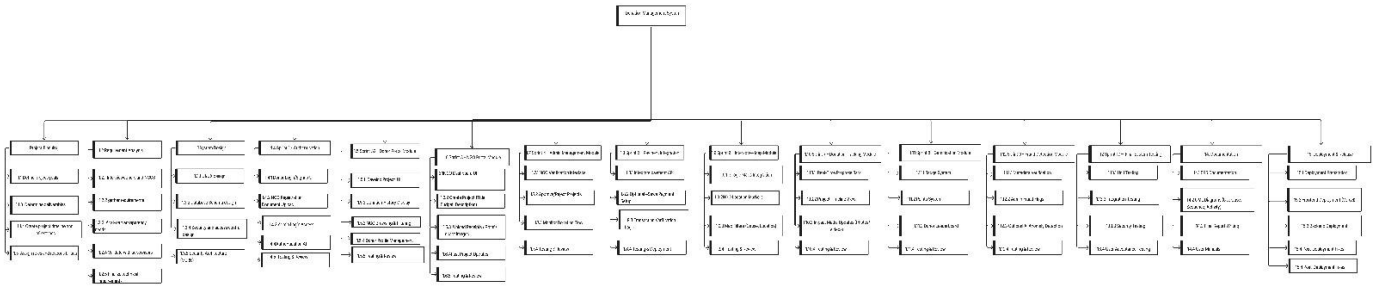

Figure 5 WBS

3.3 Gantt Chart

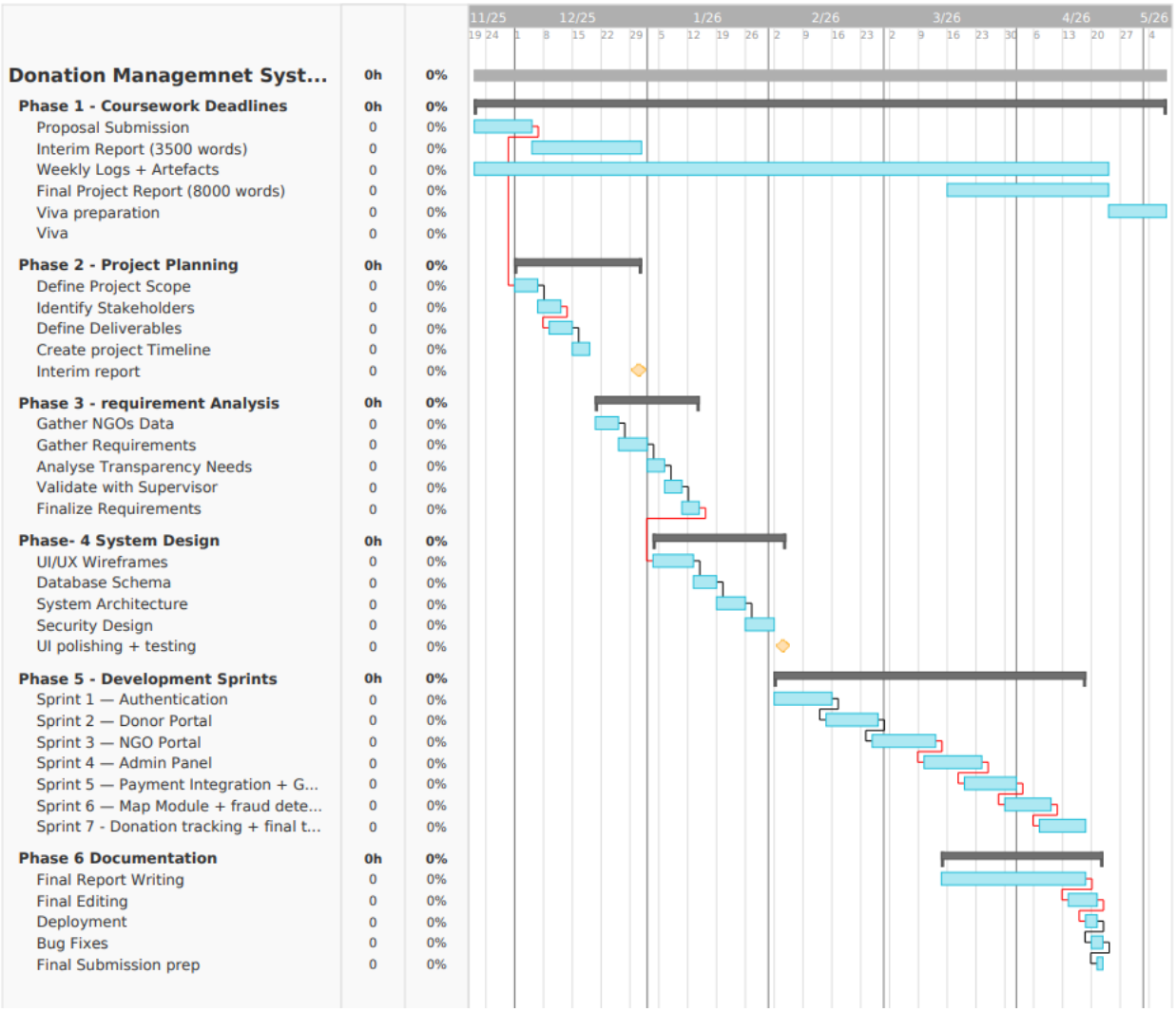

Figure 6 Gantt Chart

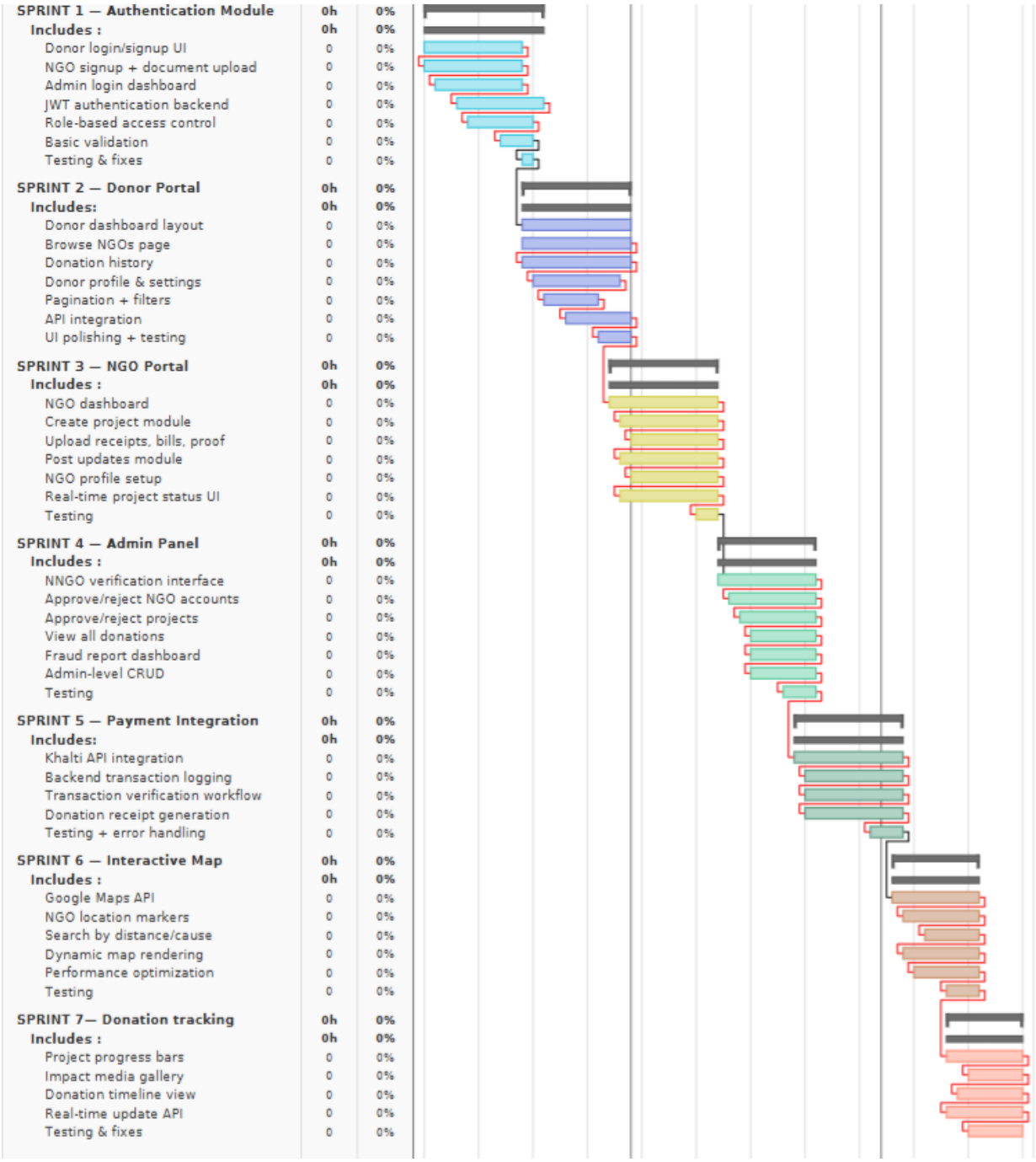

Figure 7 Gantt Chart

3.4 Requirement Gathering

The requirement gathering of the AidFlow project started with the analysis of the significant problems in Nepal that the donors and NGOs experience nowadays, including the absence of transparency, irregular updates, and inability to check the credibility of organisations. The study of international donation sites was used to find the gaps within the Nepal setting and formed the main characteristics of an exchangeable system. To narrow down the requirements, unofficial interviews and web surveys were performed with the potential donors and students, which also gave some information on the expectations, including real-time updates, secure payments, and the ability to see the impact. These results, which are presented in the appendix, proved that it was necessary to have a single platform such as AidFlow that would simplify the donations process and enhance the confidence between the donor and the NGOs.

For more Survey Form Details click here: Survey Form Screenshot

4. Development to Date

The development of AidFlow began with in depth discussions to clearly outline the scope of the project, the core problems of the contribution environment in Nepal, and the purpose of a platform of open equal opportunities and trust. The Agile Scrum technique informed the process and employed short sprints to give the process flexibility and consistency in its development. Early stages that involved mapping system functionalities, drawing out UML diagrams including use case, sequence, activity, and ERD, and wireframe the donor and NGO interfaces were more related with system planning and not code. These stages of design provided a strong basis of development.

4.1 System Architecture Design

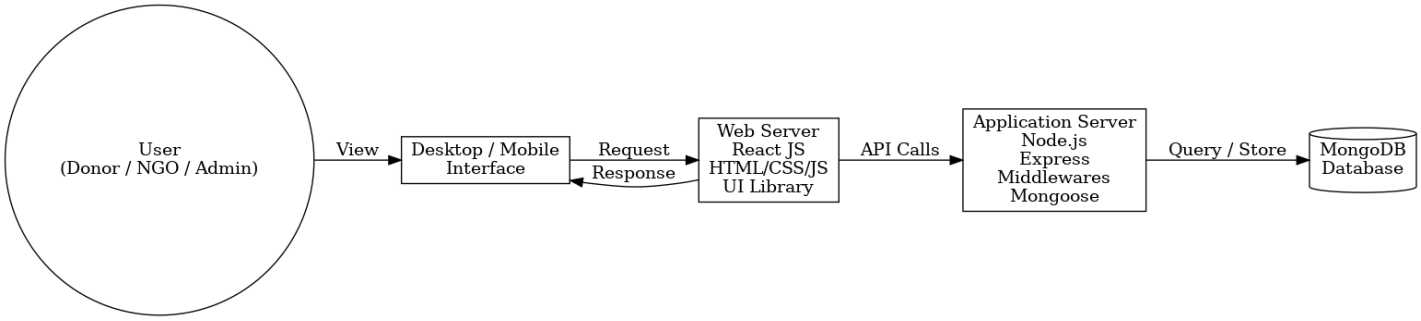

Figure 8 System Architecture Design

4.2 Use Case Diagram

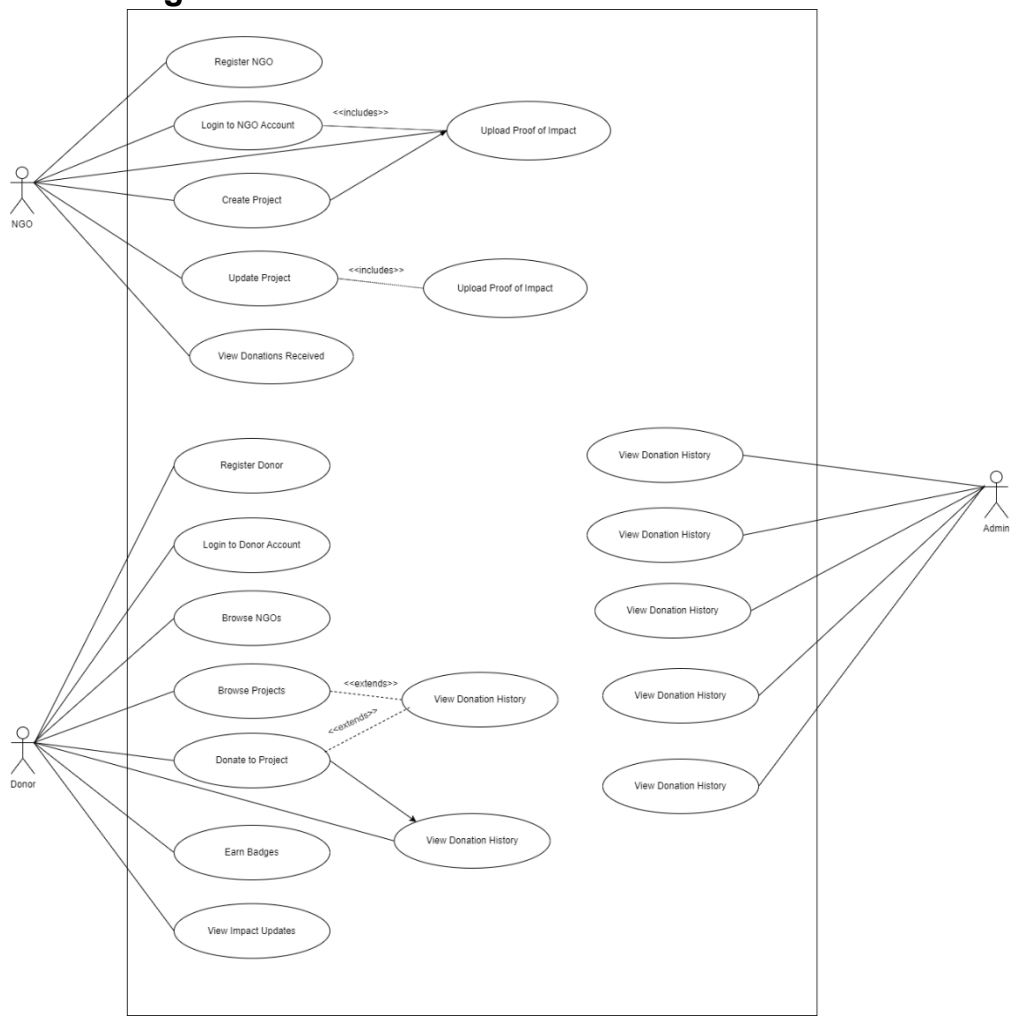

Figure 9 Use Case Diagram

For [High Level and Expanded Level Descriptions]

4.3 Data Flow Diagram

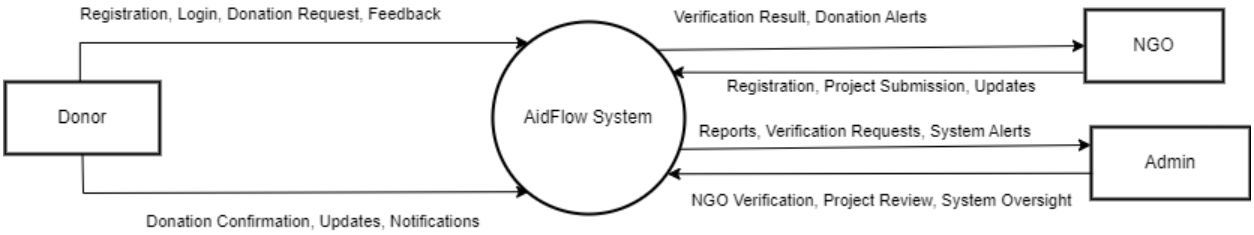

Figure 10 Data Flow Diagram level 0

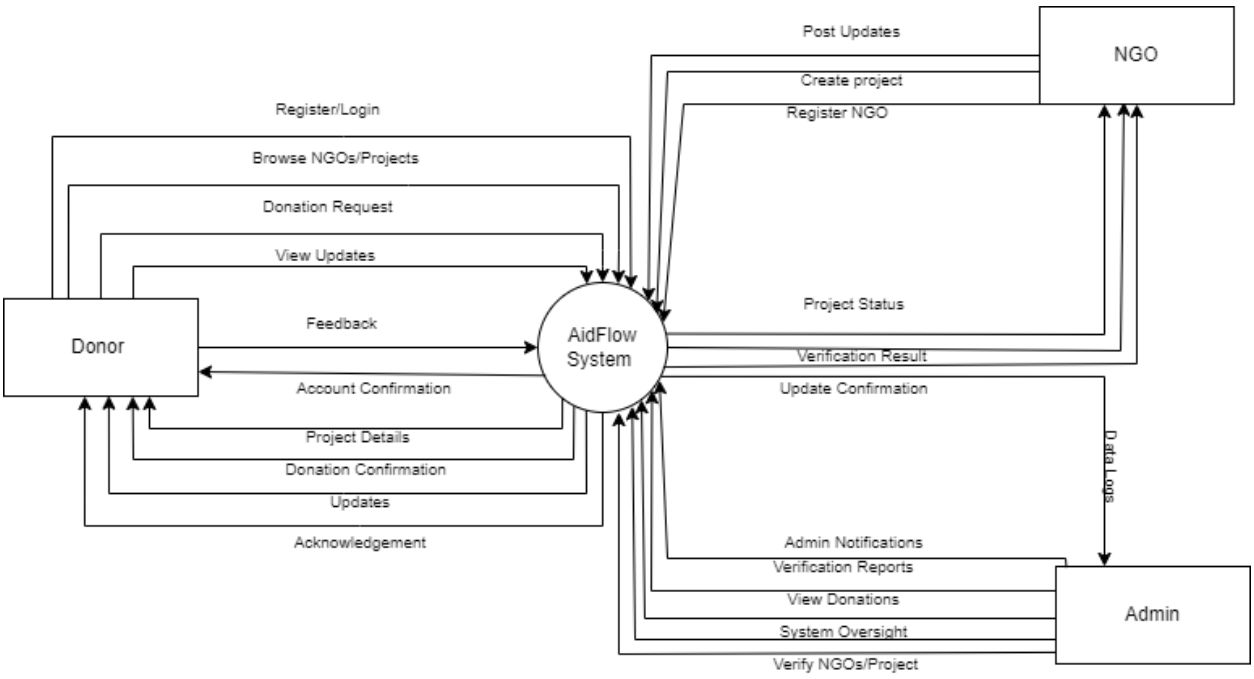

Figure 11 Data Flow Diagram Level 1

4.4 Class diagram

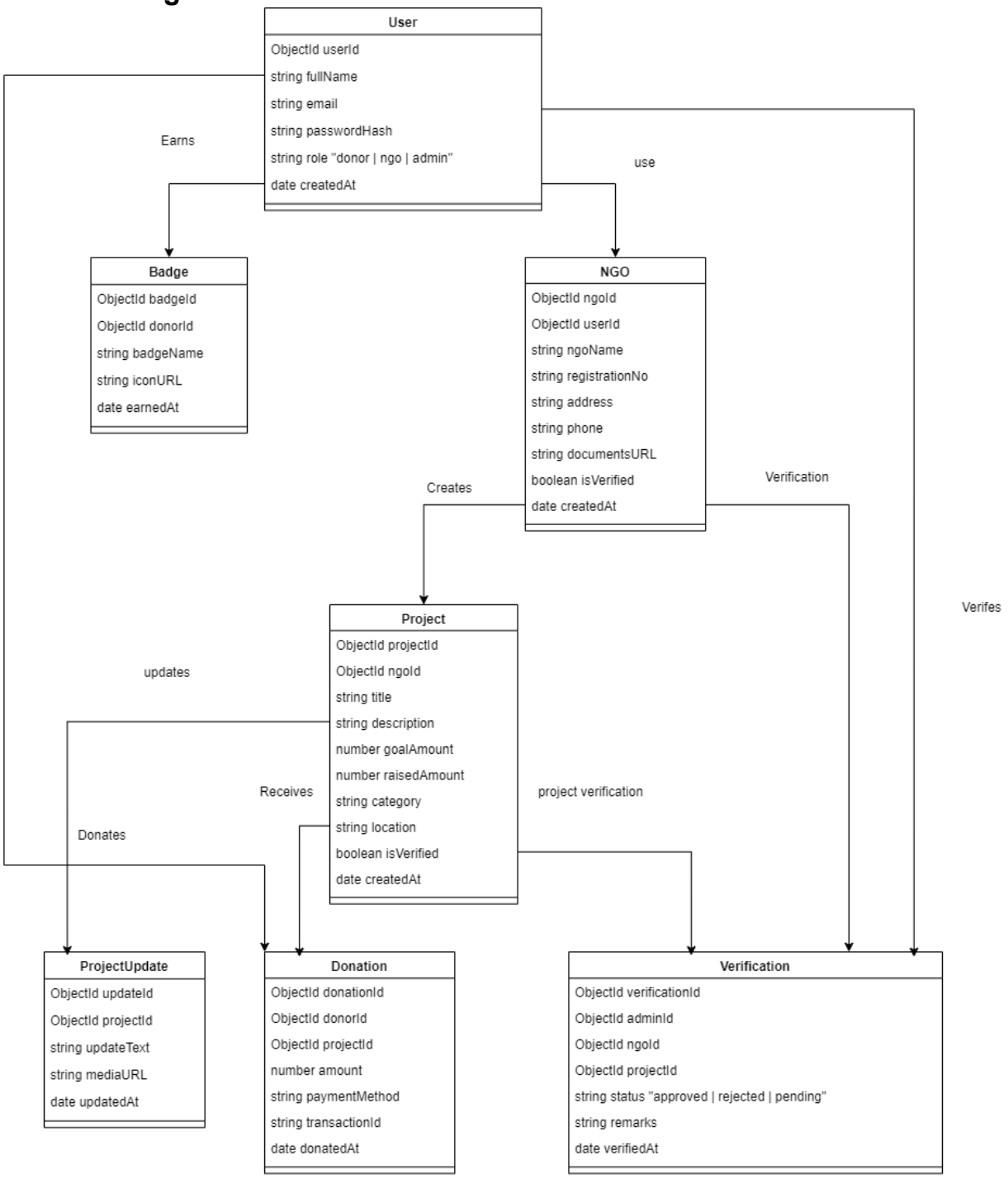

Figure 12 Class diagram

4.5 Entity Relationship Diagram

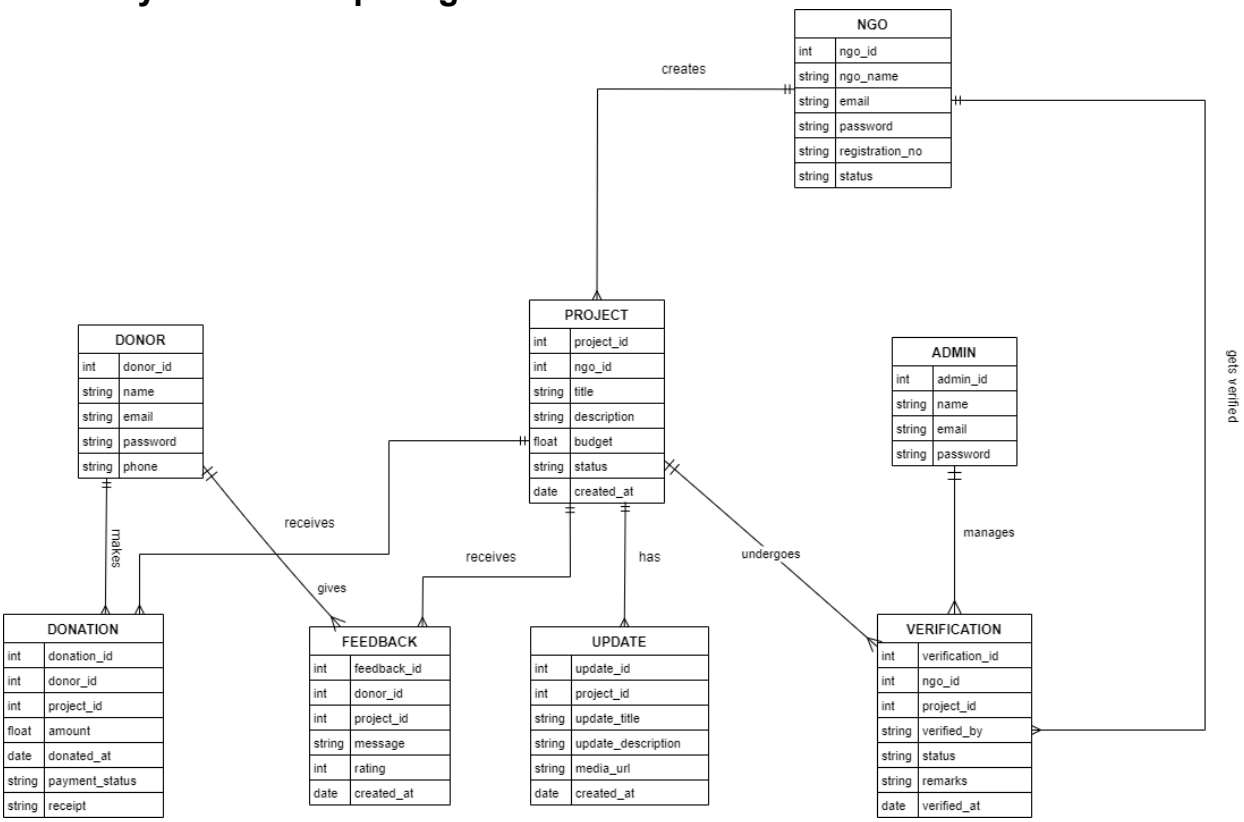

Figure 13 Entity Relationship Diagram

4.6 Sequence Diagram

4.6.1 Sequence diagram of Login

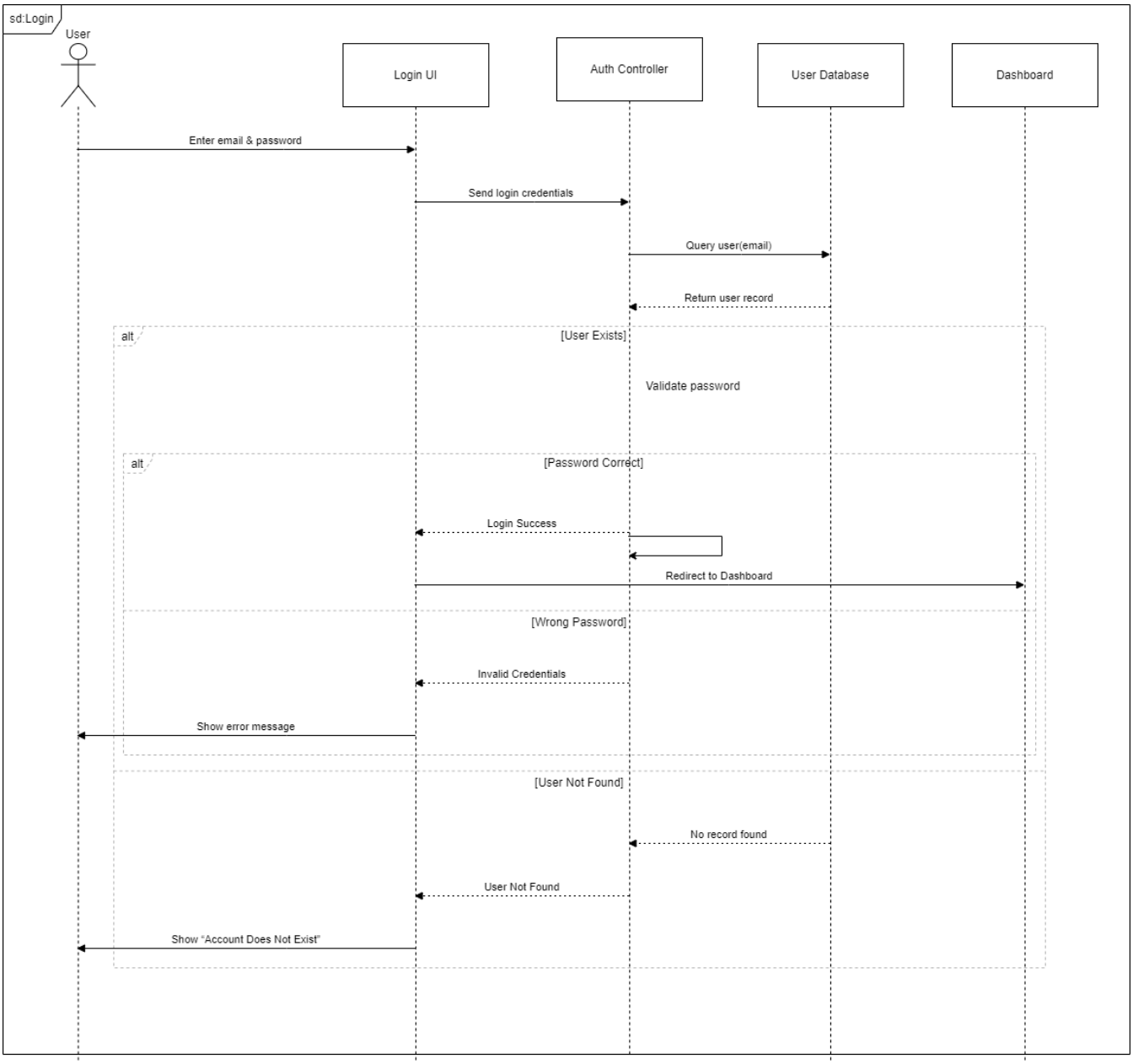

Figure 14 Sequence diagram of Login

4.6.2 Sequence diagram of NGO Registration

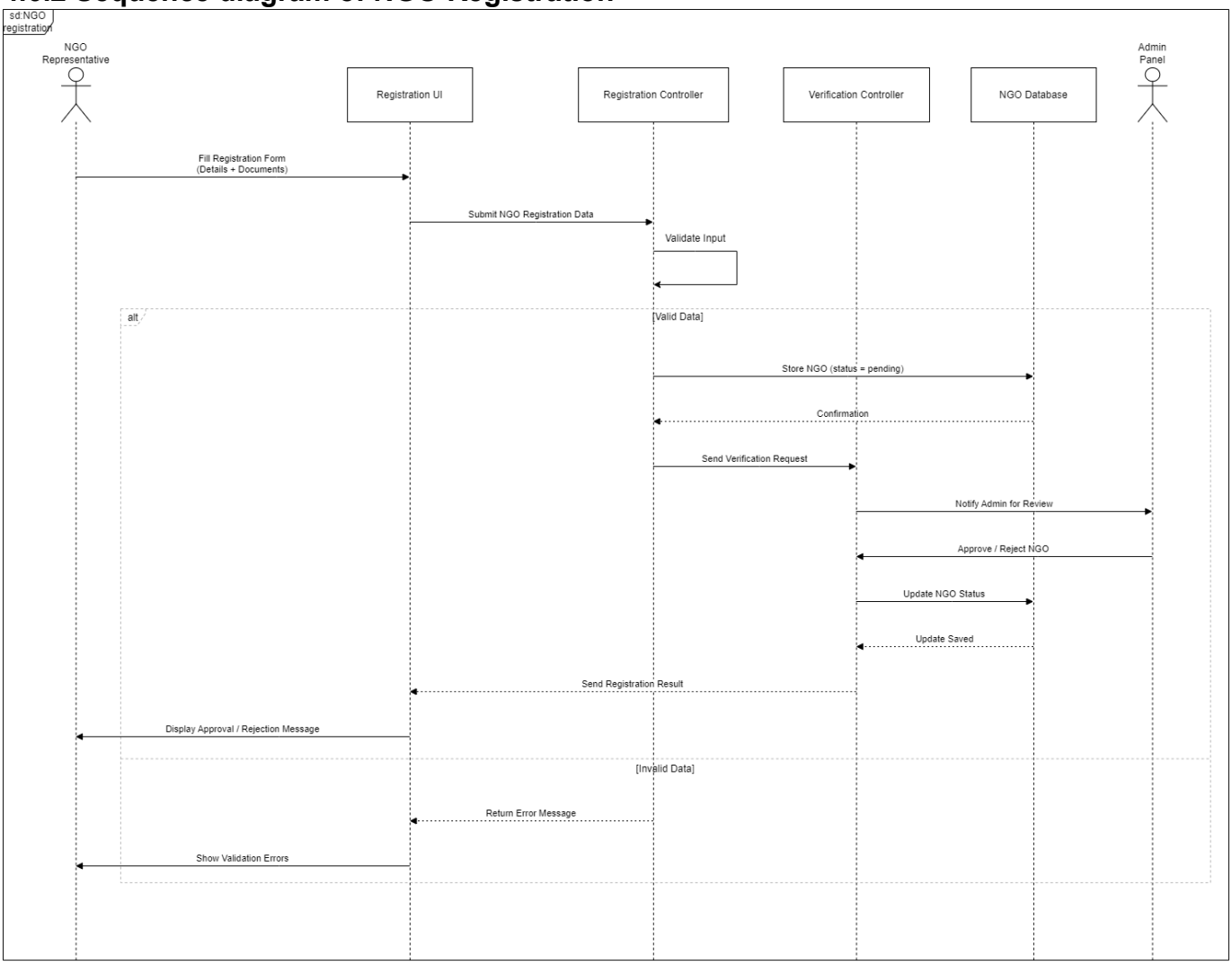

Figure 15 Sequence diagram of NGO Registration

4.6.3 Sequence diagram of Donation Process

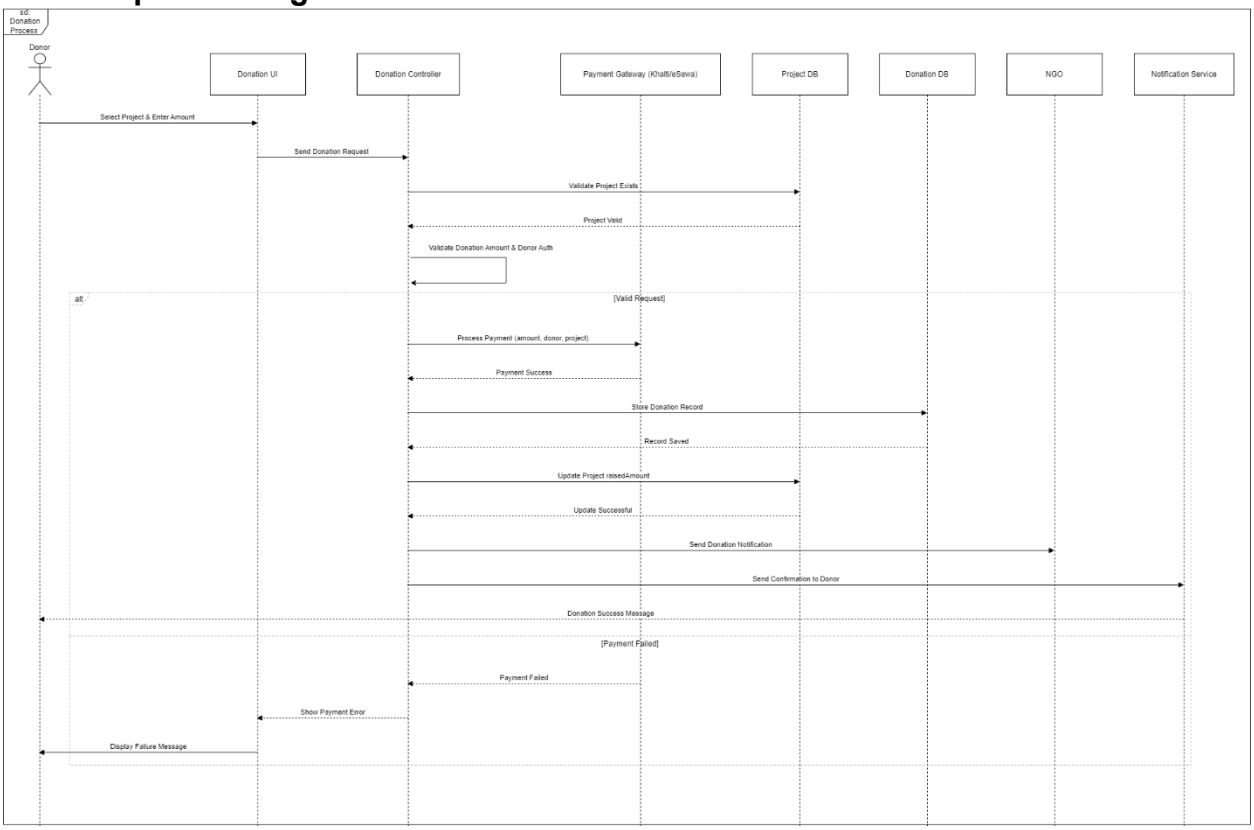

Figure 16 Sequence diagram of Donation Process

4.6.4 Sequence diagram of Project Update

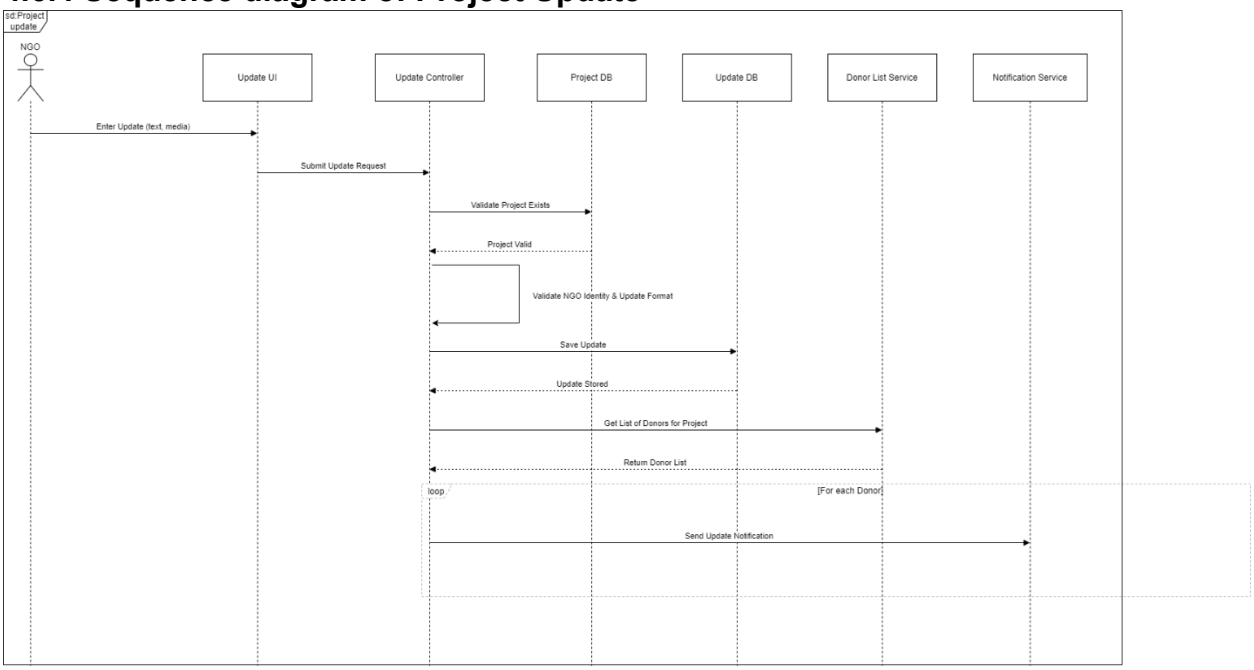

Figure 17 Sequence diagram of Project Update

4.7. Flow chart

4.7.1 Flowchart of Login

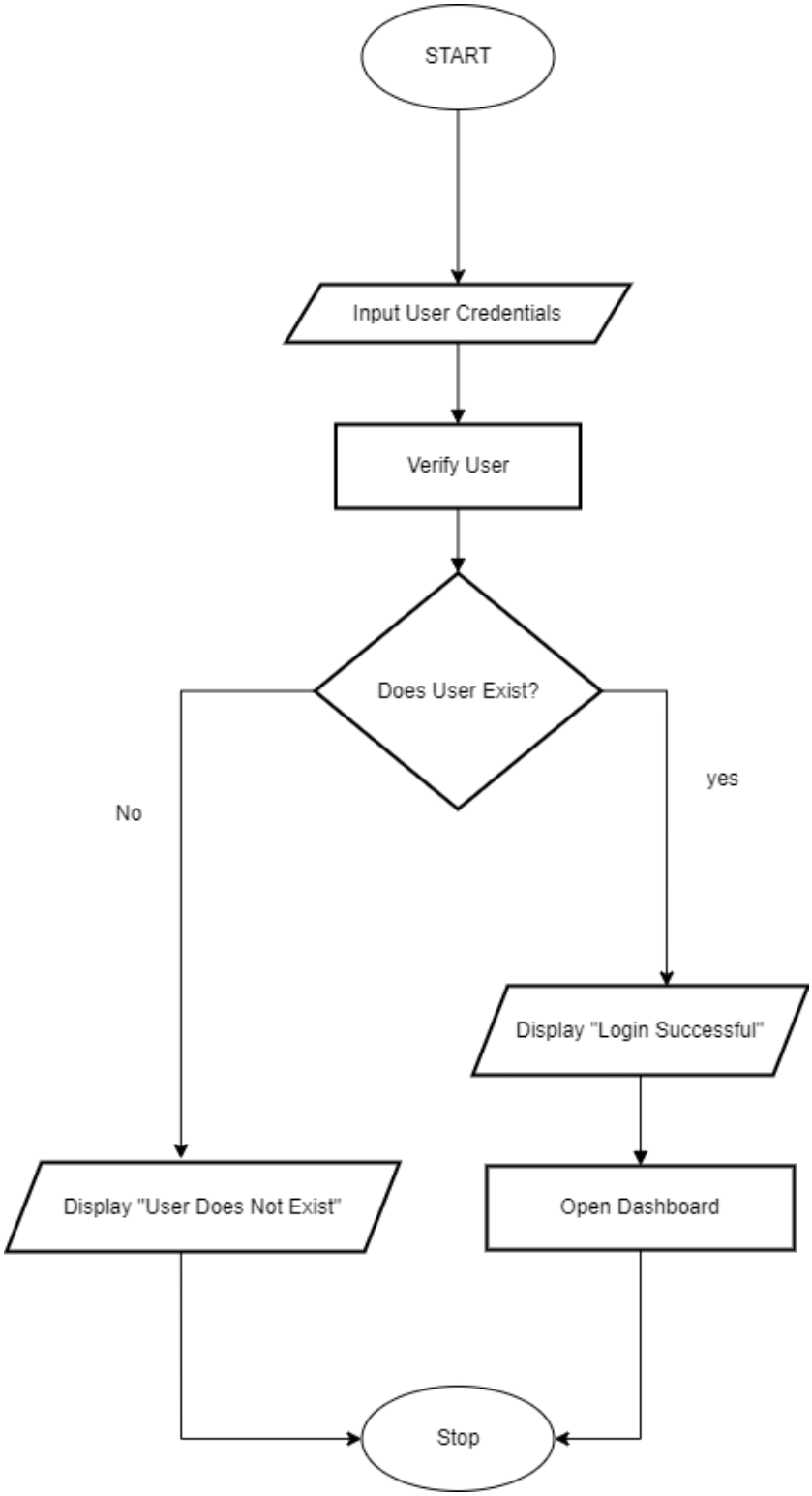

Figure 18 Flowchart of Login

4.7.2 Flowchart of NGO registration

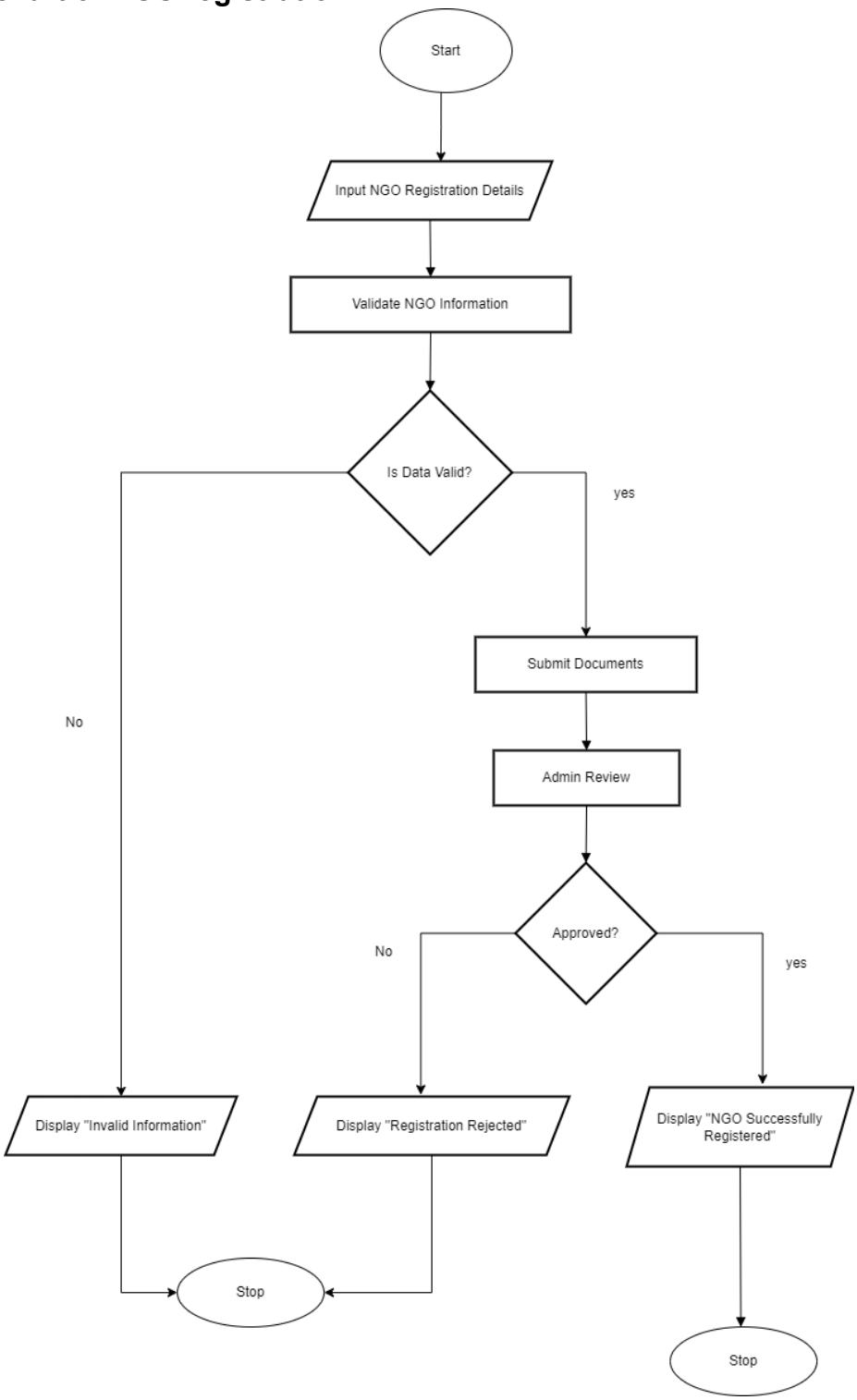

Figure 19 Flowchart of NGO registration

4.7.3 Flowchart of Donation Process

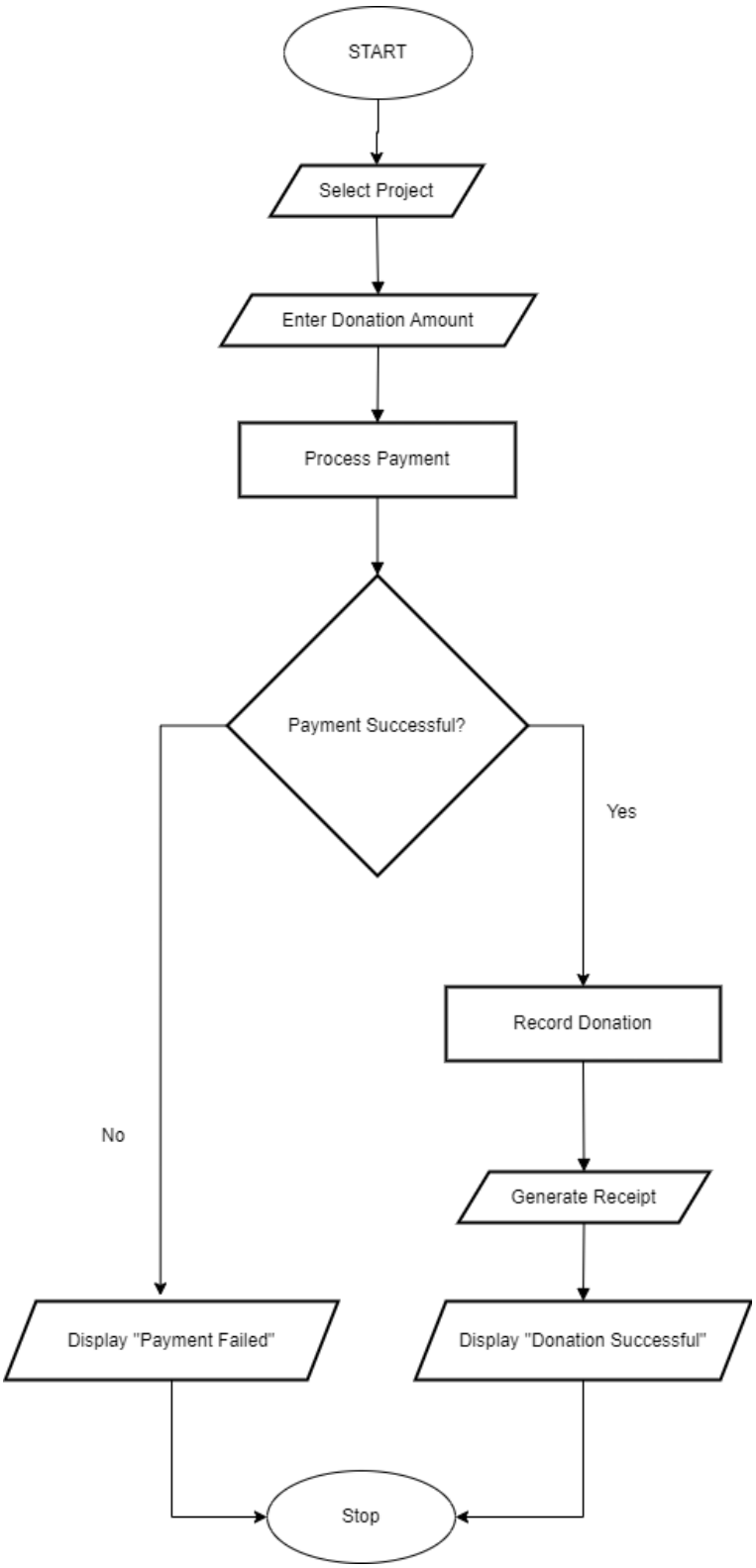

Figure 20 Flowchart of Donation Process

4.7.4 Flowchart of Project Update

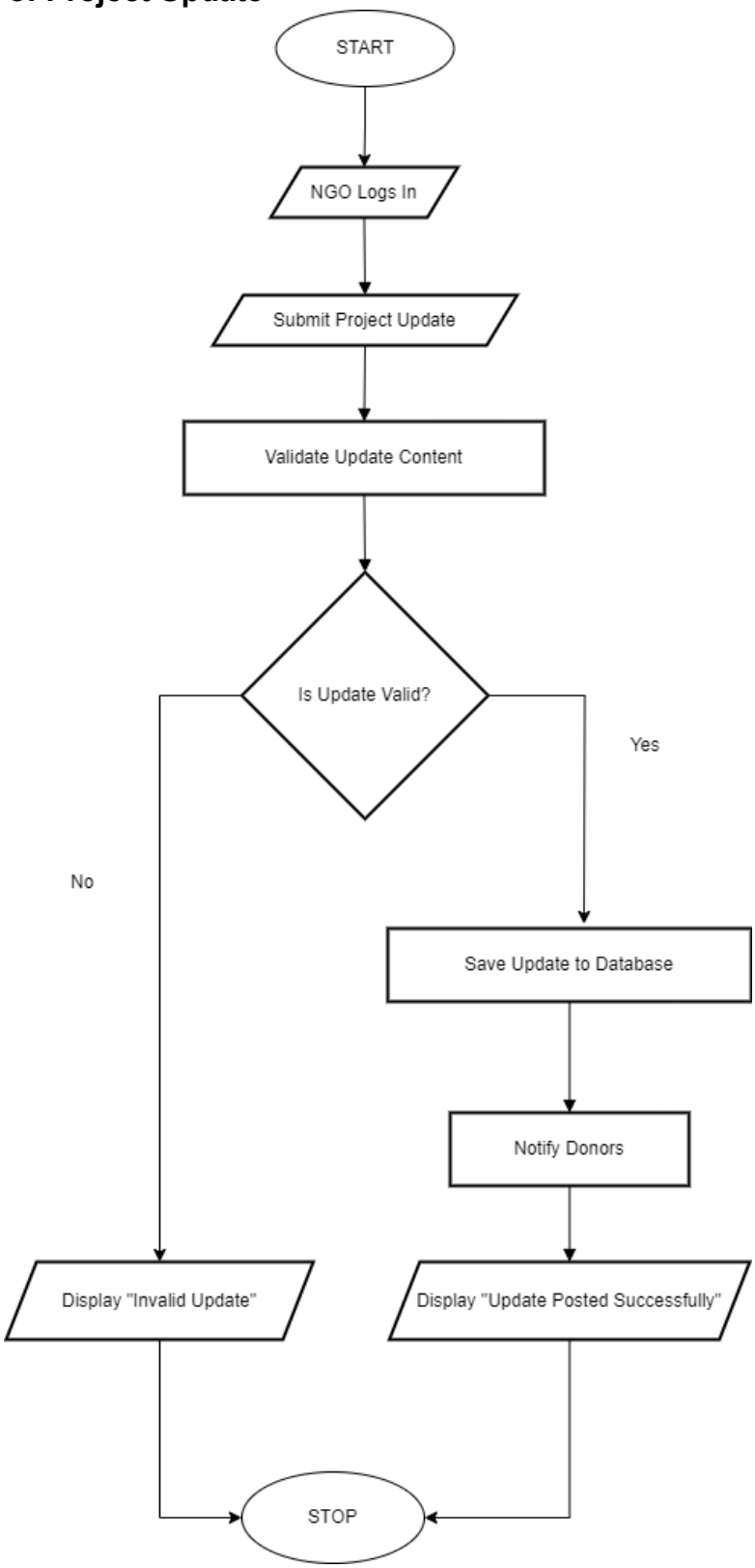

Figure 21 Flowchart of Project Update

4.8 Wire Frame

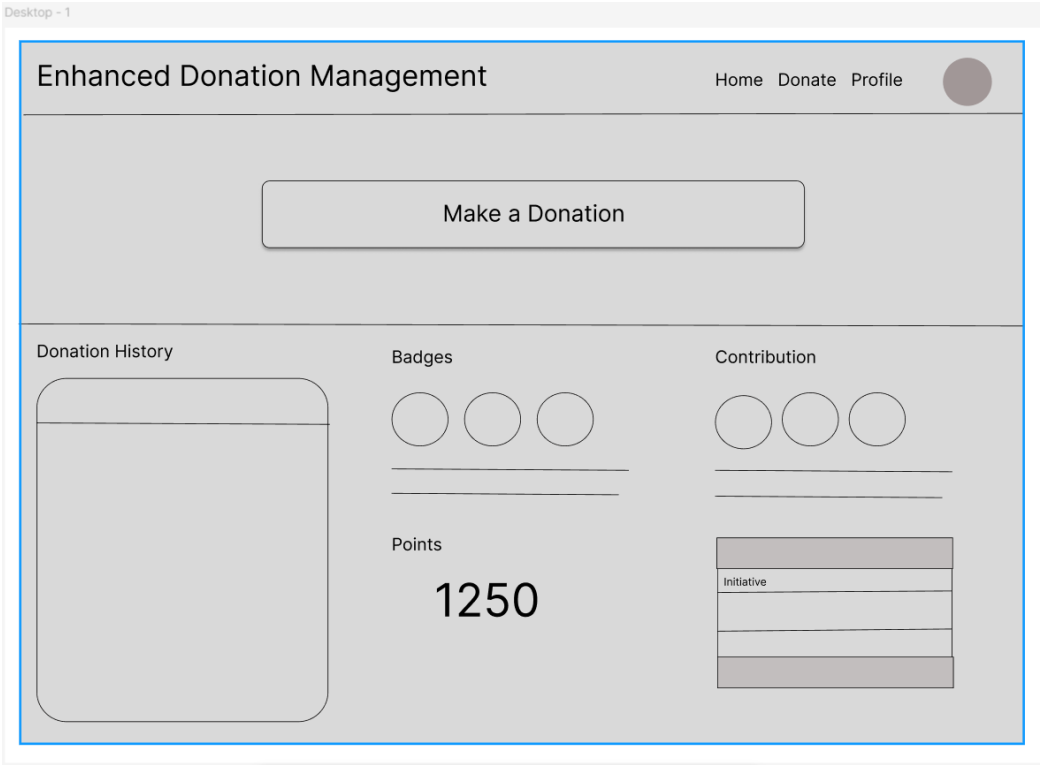

Figure 22 Wireframe home page

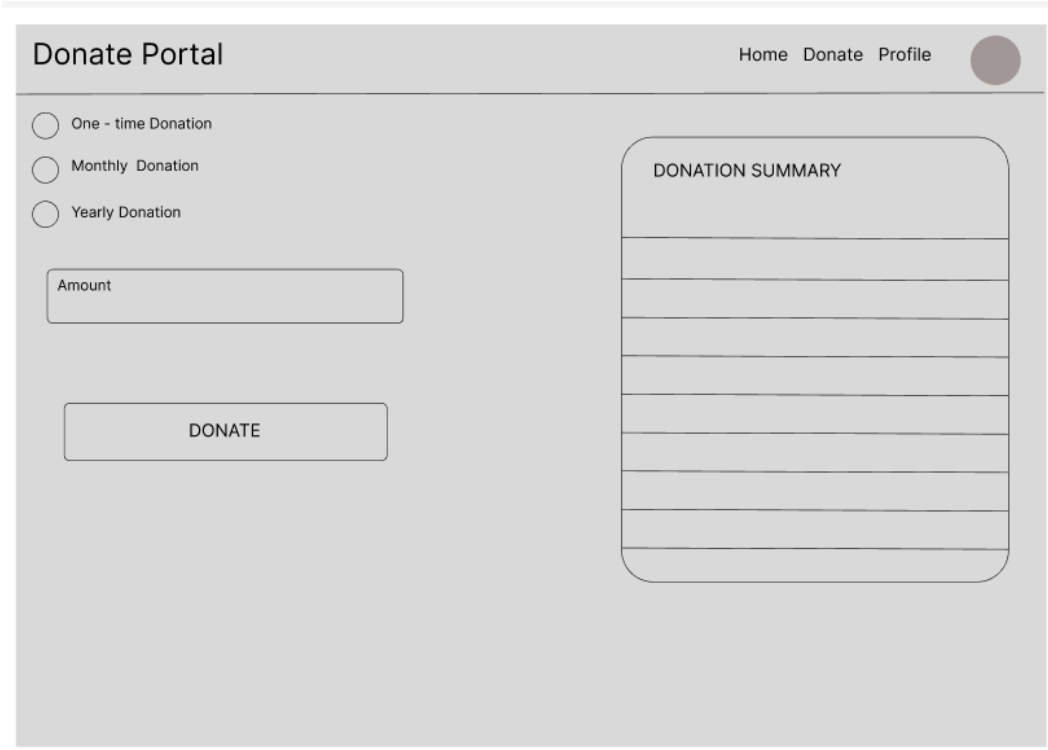

Figure 23 Donate Portal

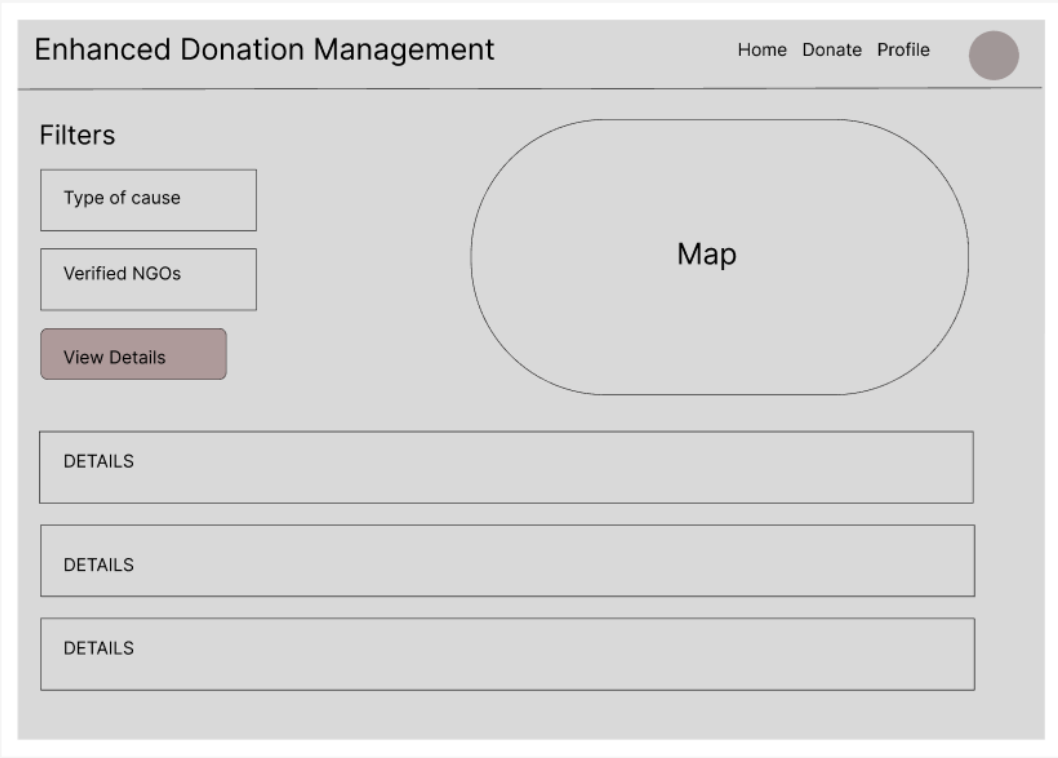

Figure 24 Enhanced Donation Management

Figure 25 NGO Verification

4.9 UI Design

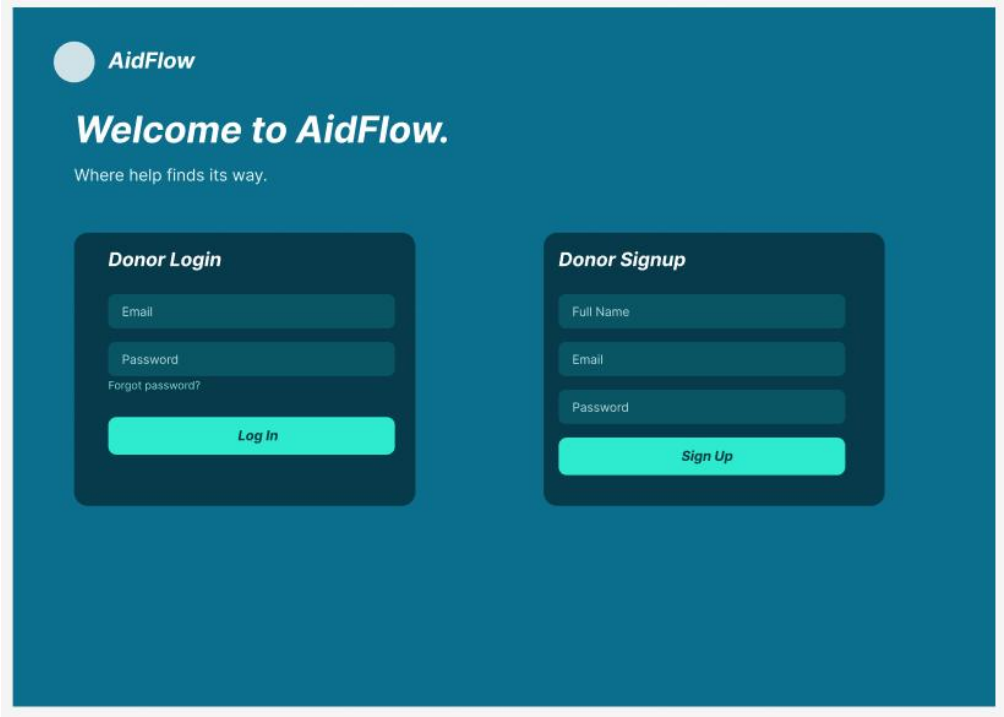

Figure 26 UI Login

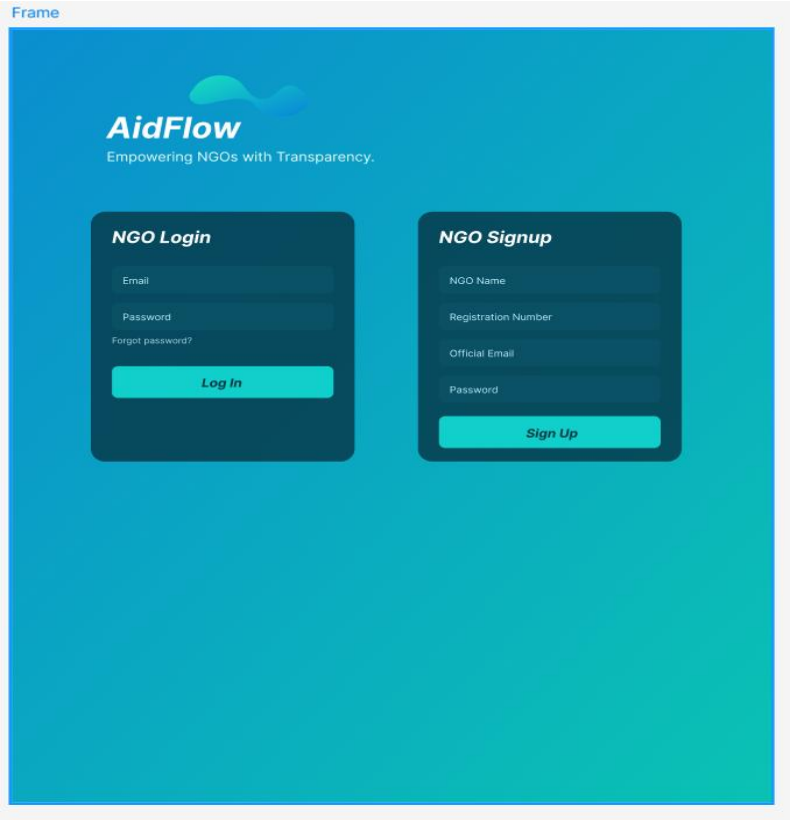

Figure 27 UI Login

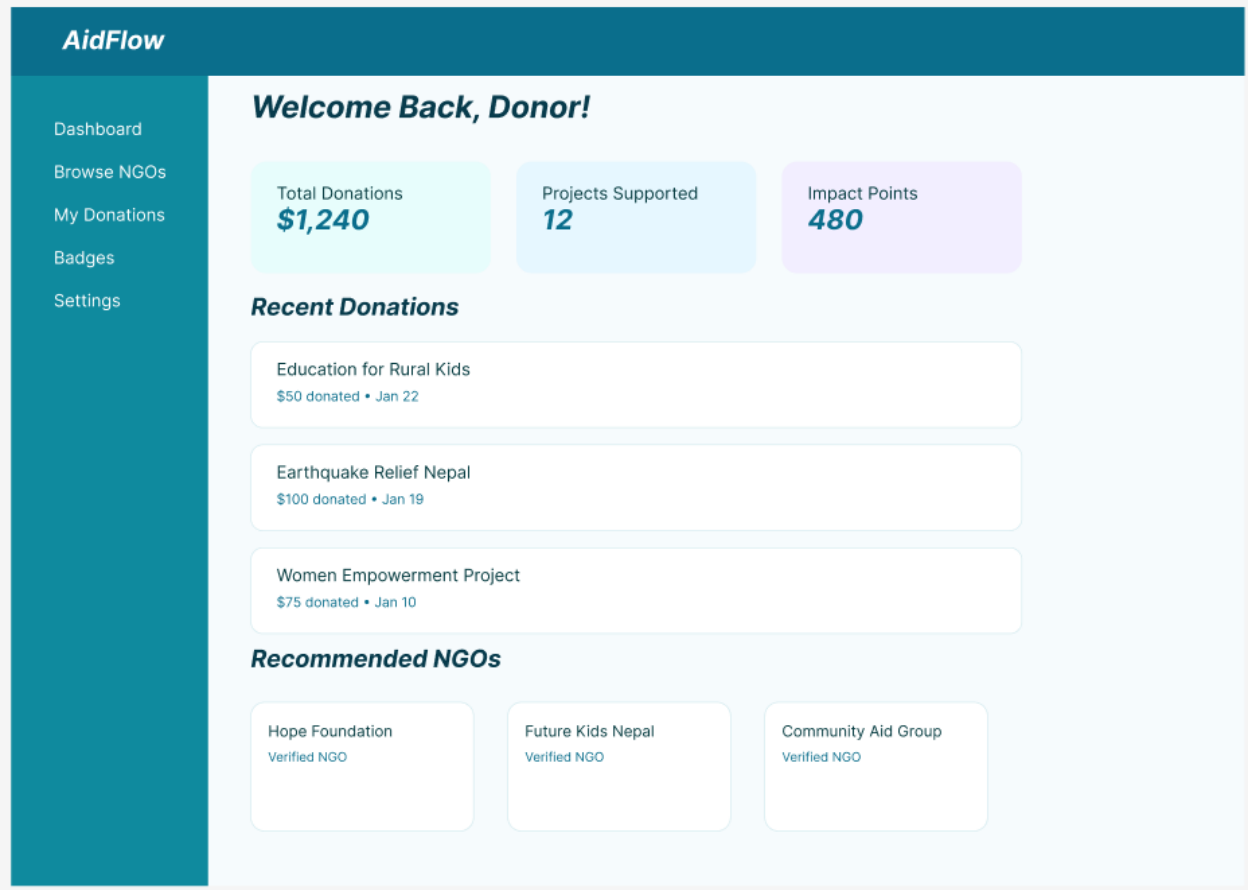

Figure 28 AidFlow Home page

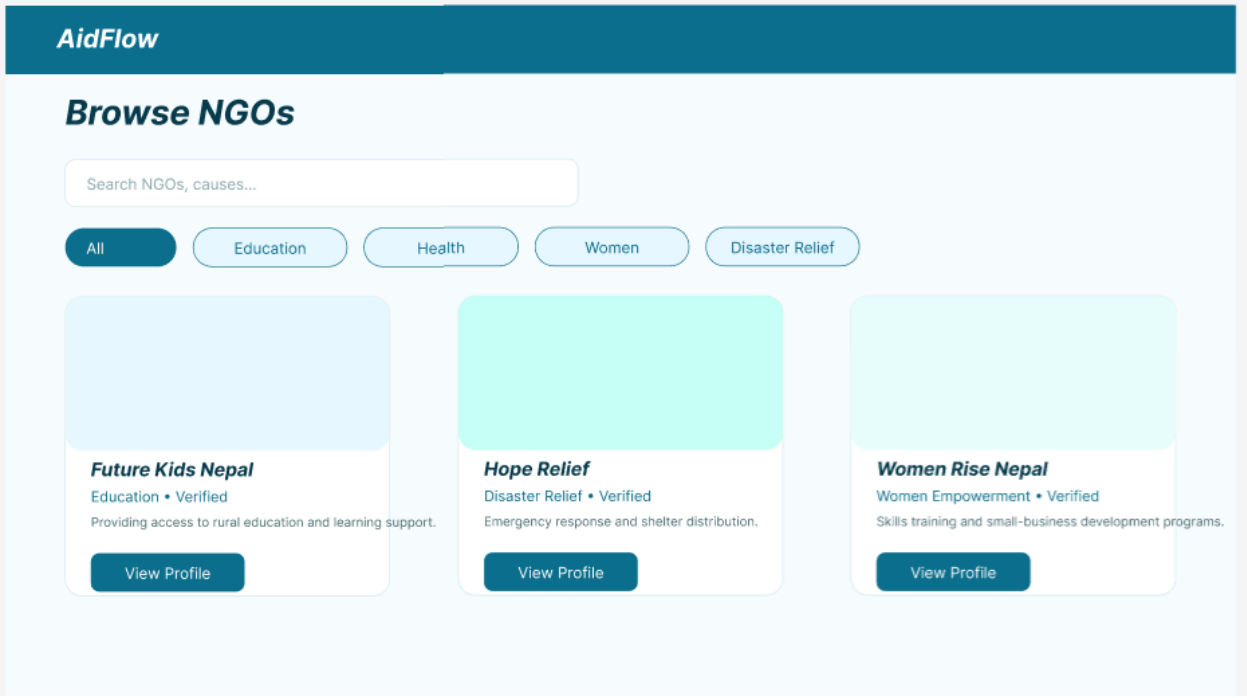

Figure 29 AidFlow Browse page

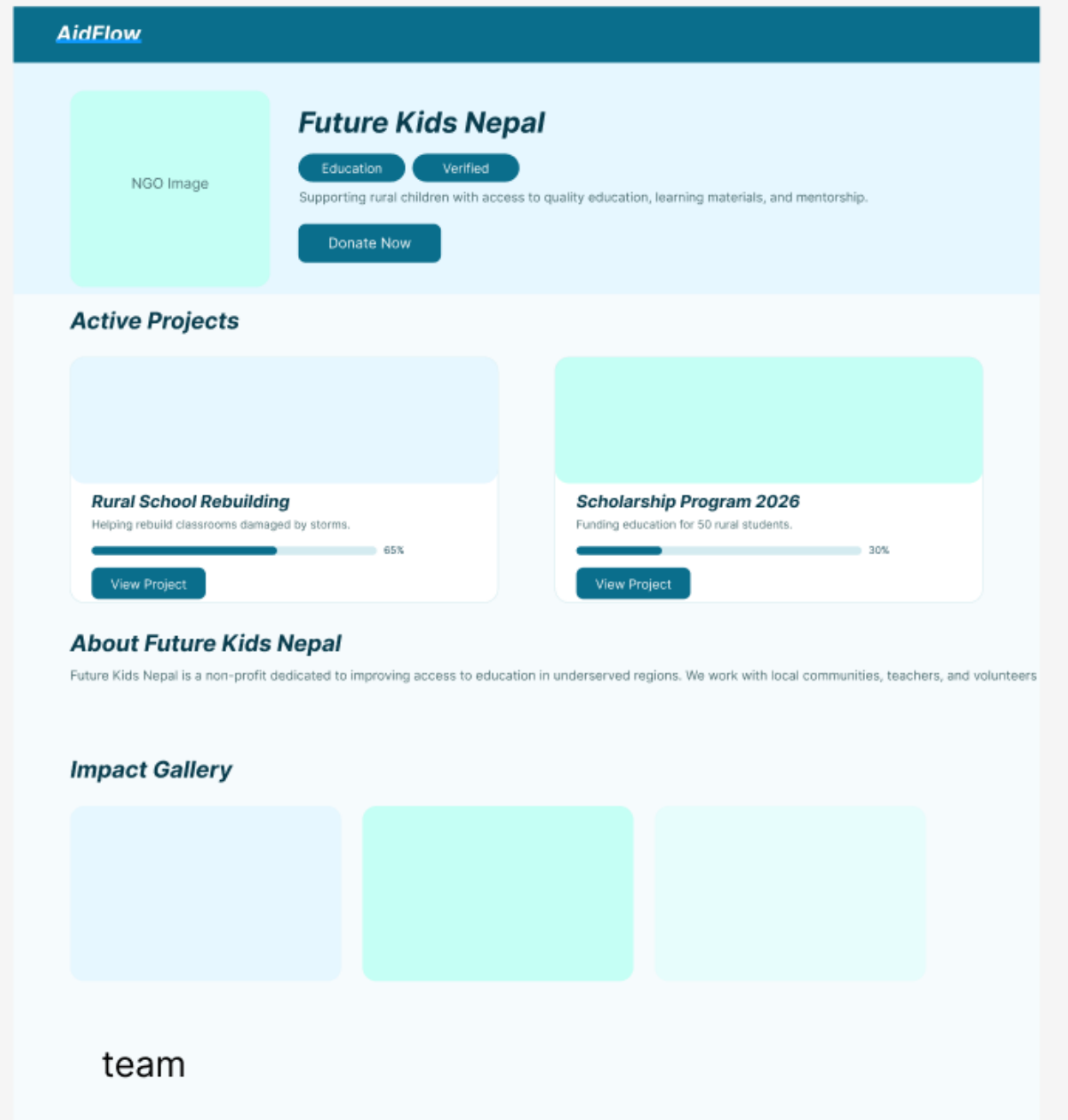

Figure 30 AidFlow NGO page

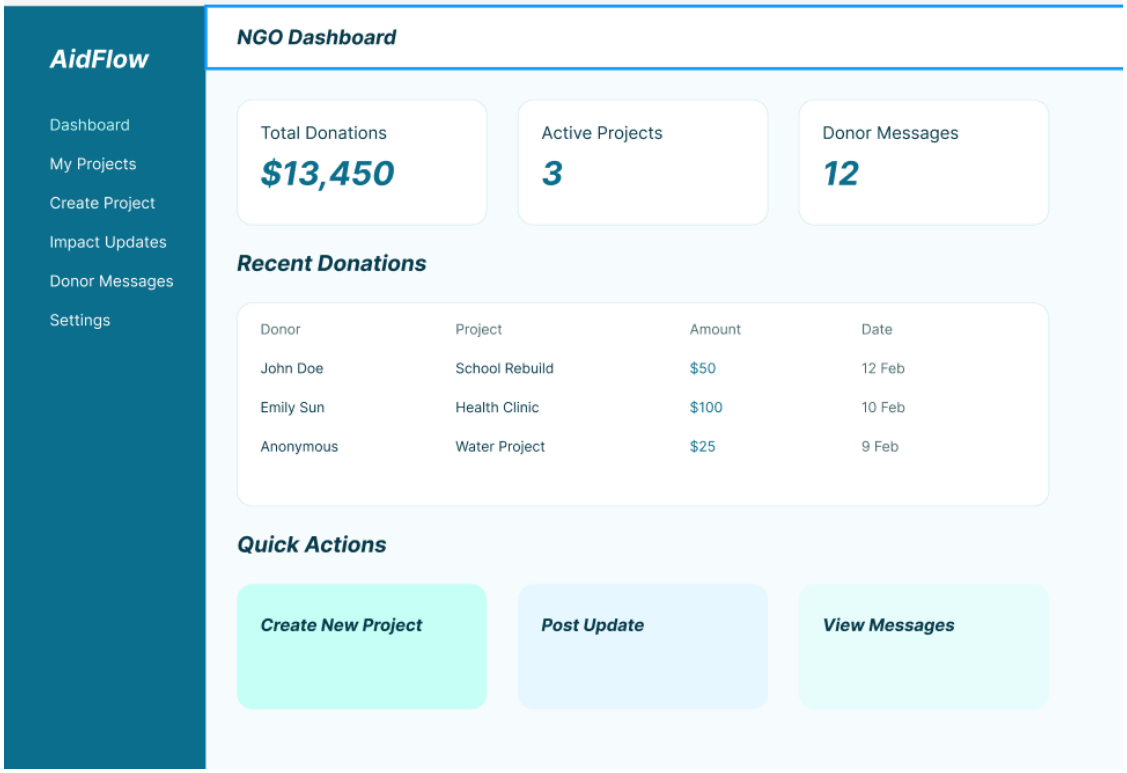

Figure 31 NGO Dashboard

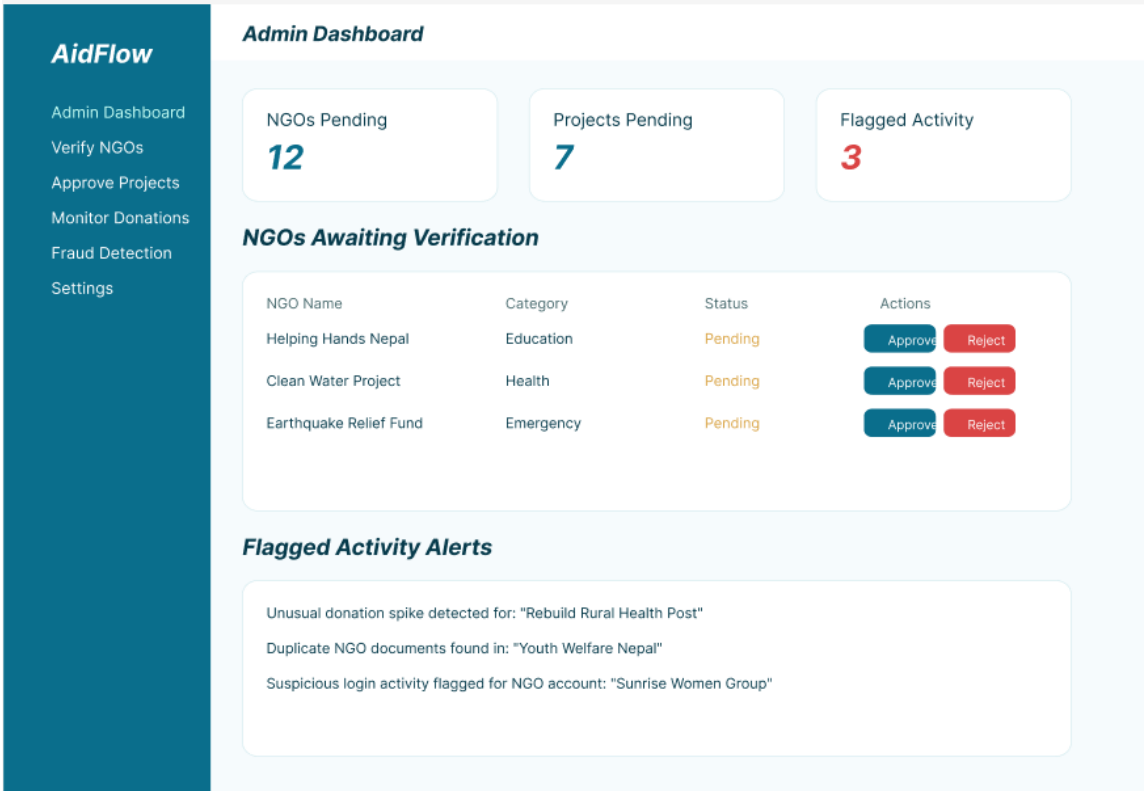

Figure 32 Admin Dashboard

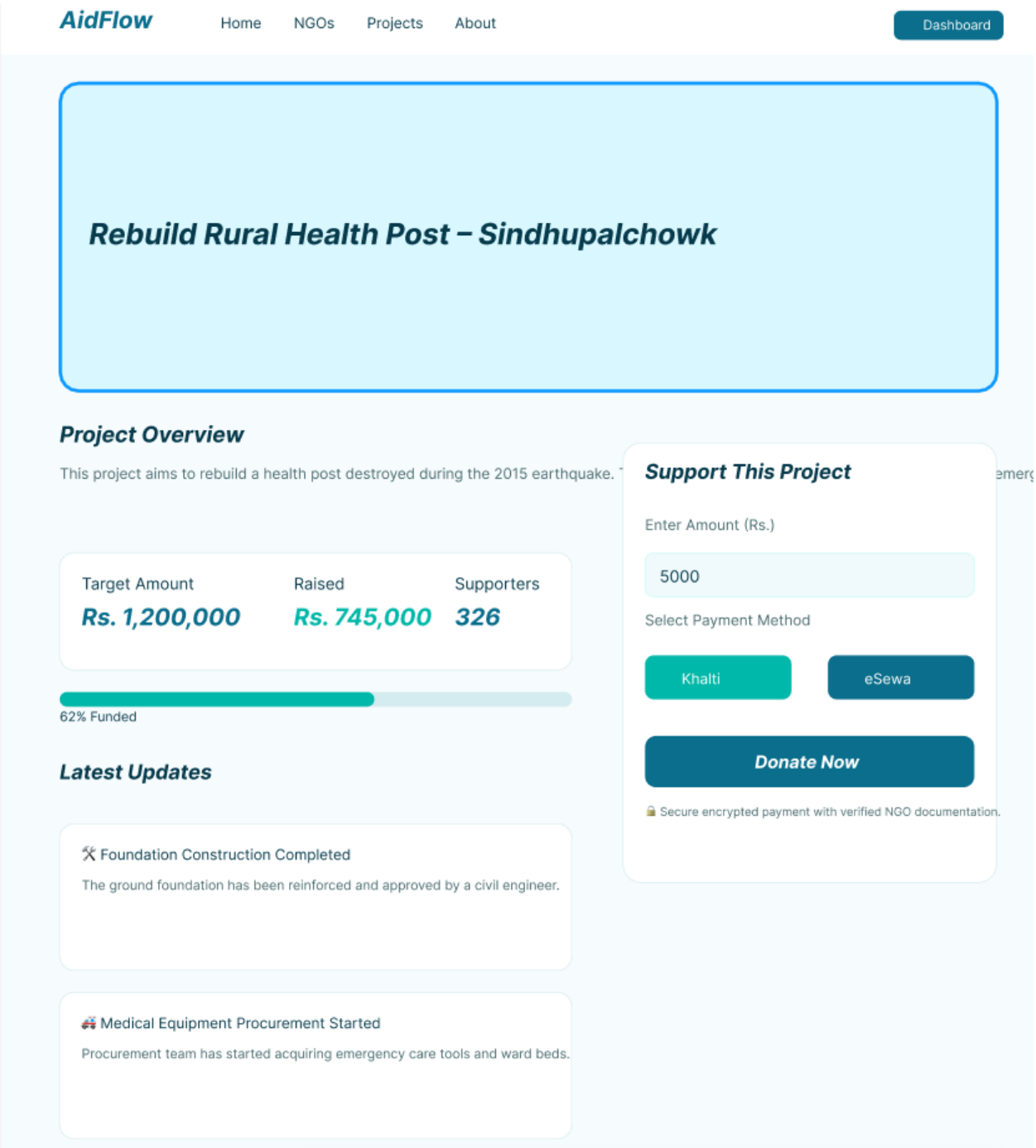

Figure 33 Aid flow project page

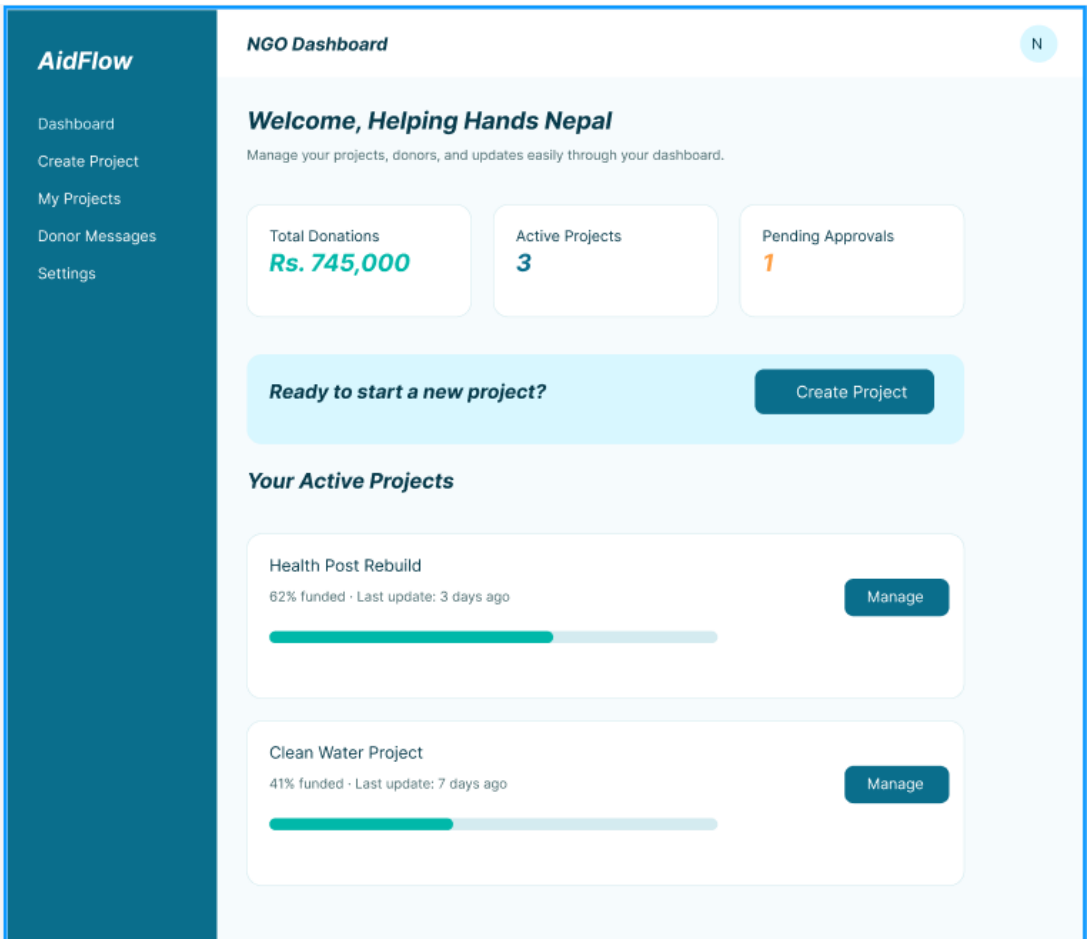

Figure 34 NGO Dashboard

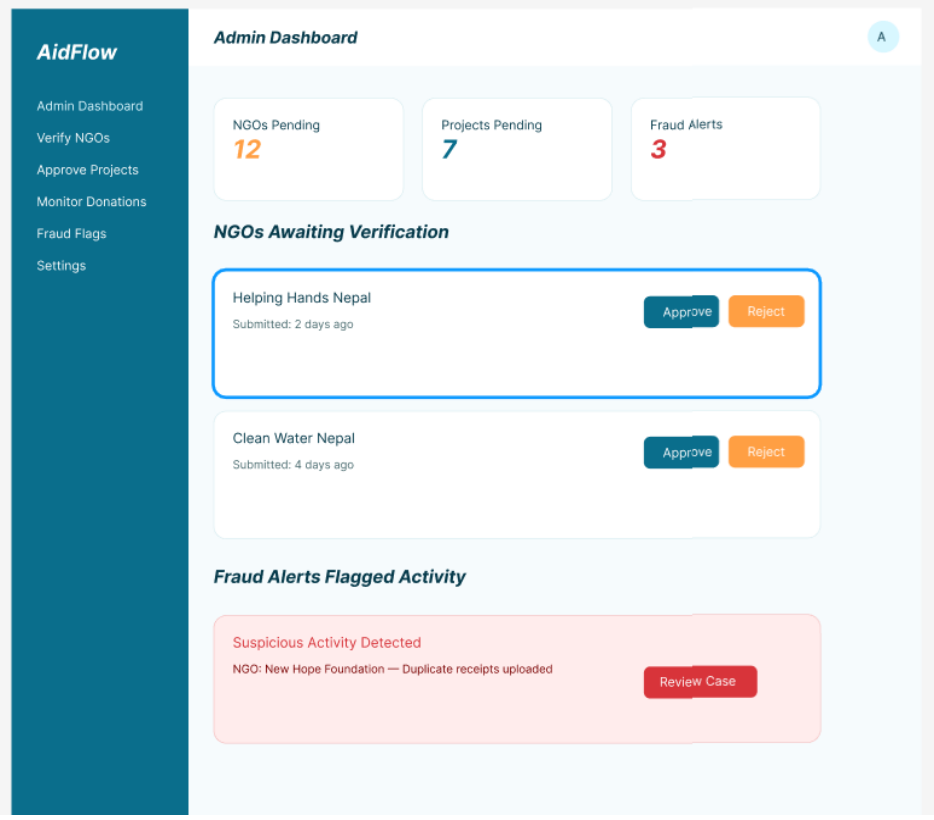

Figure 35 Admin Dashboard

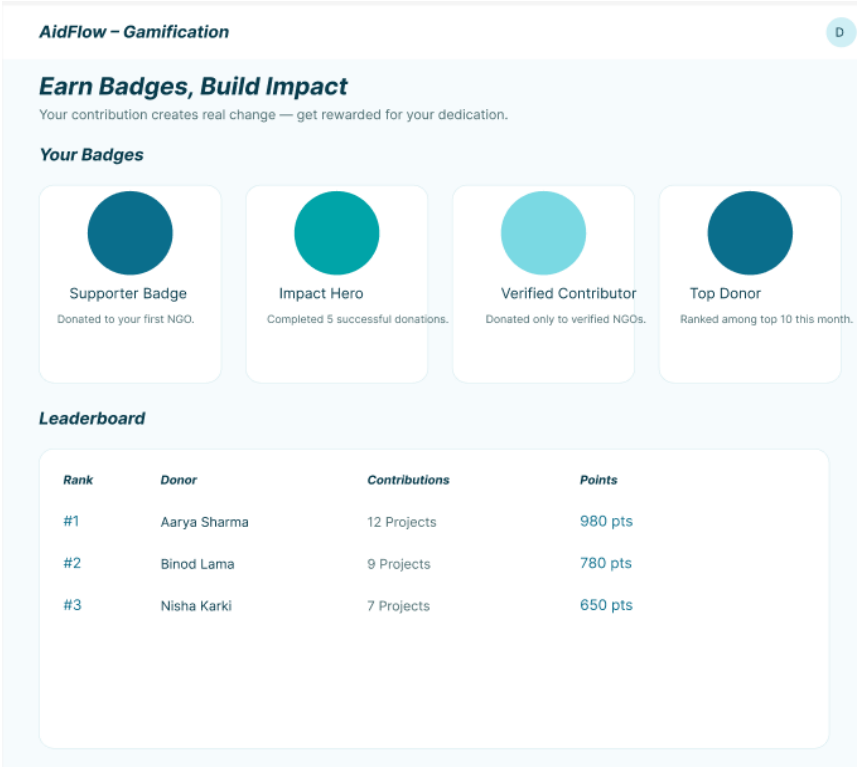

Figure 36 badge Showcase

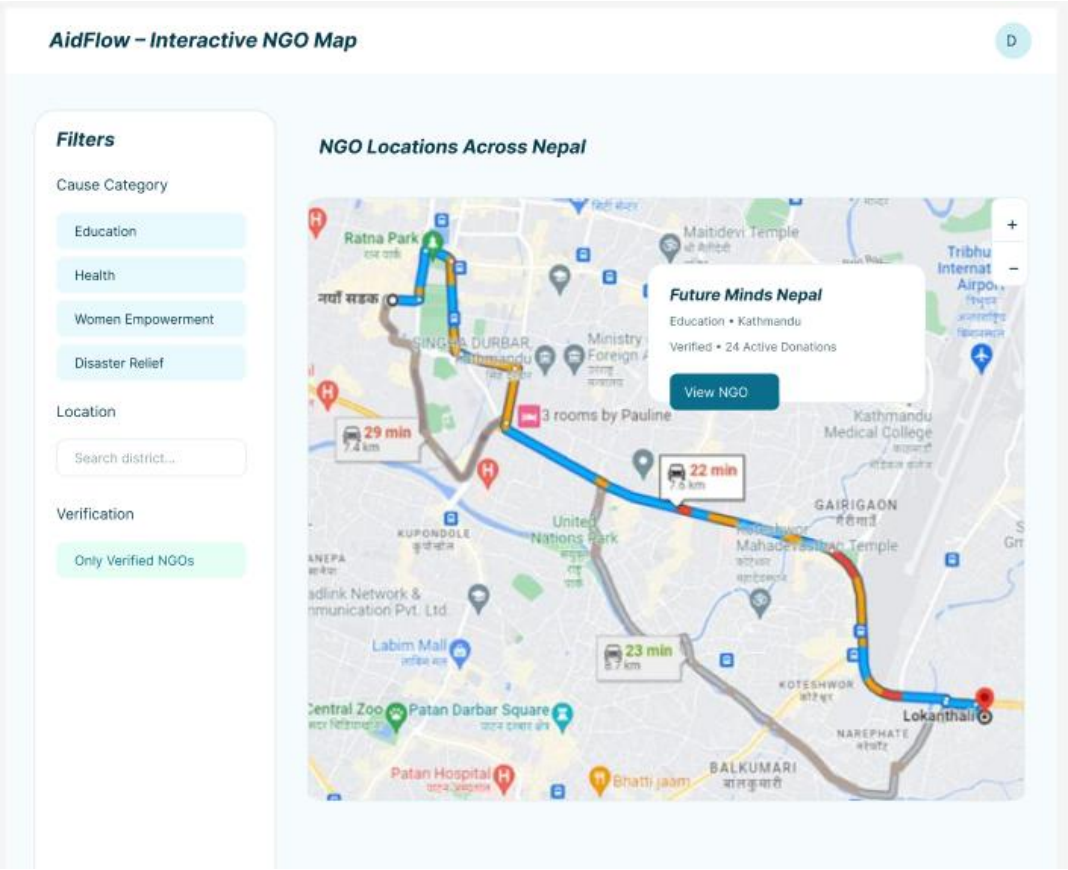

Figure 37 Map Integration

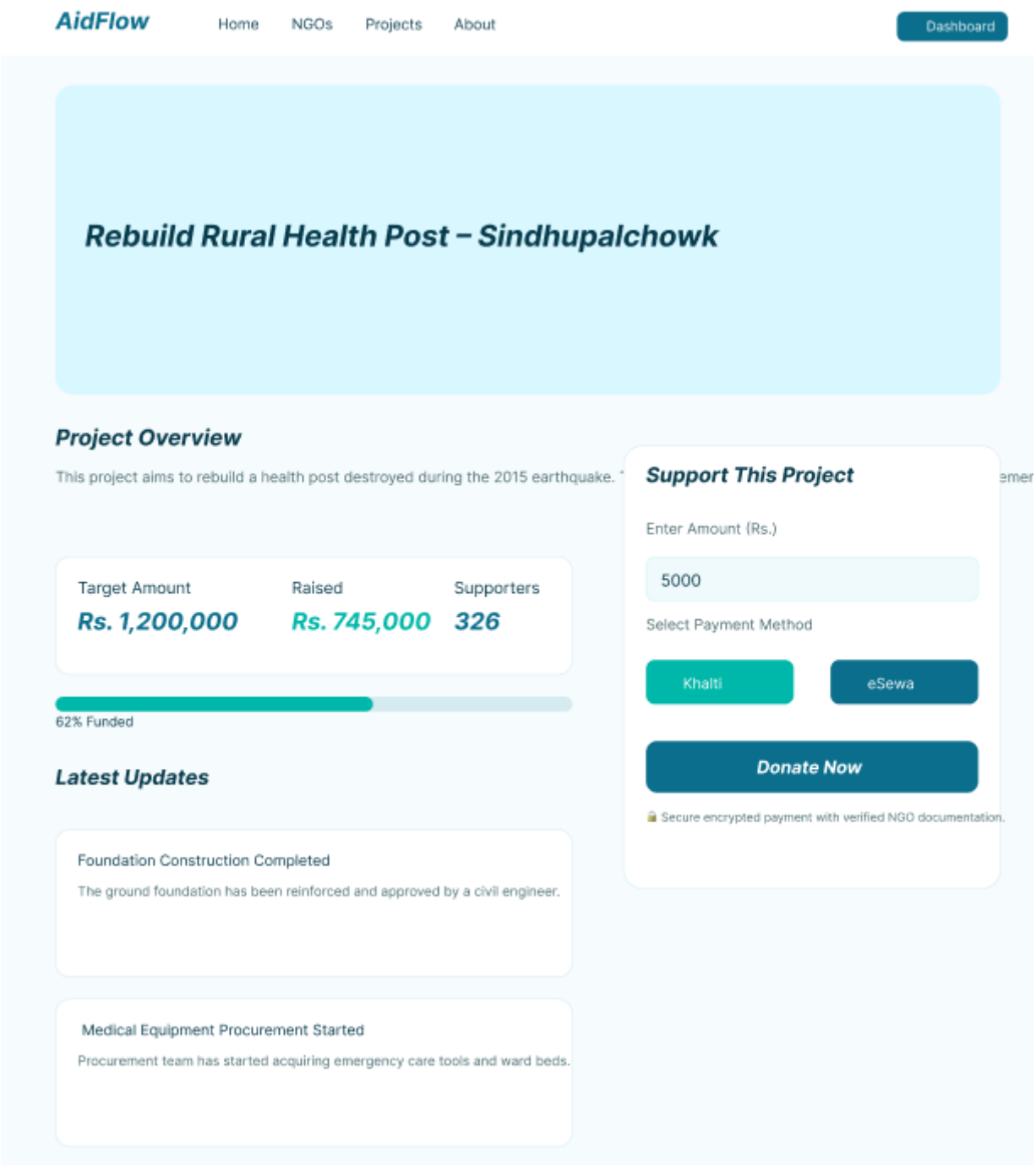

Figure 38 NGO Home Page

4.10 Development status with Evidence

```
import { useState, ChangeEvent, FormEvent } from "react";
import "./aidflow-login.css";
import { login } from "../../shared/config/api";
import { AxiosResponse, AxiosError } from "axios";
import { toast } from "react-toastify";
import "react-toastify/dist/ReactToastify.css";
import { useNavigate } from "react-router-dom";

export default function Login() {
  const navigate = useNavigate();

  const [loginData, setLoginData] = useState({
    email: "",
    password: "",
  });

  const [signupData, setSignupData] = useState({
    ngoName: "",
    registrationNumber: "",
    email: "",
    password: "",
  });

  const [loading, setLoading] = useState(false);

  // Handle login form changes
  const handleLoginChange = (e: ChangeEvent<HTMLInputElement>) => {
    setLoginData({ ...loginData, [e.target.name]: e.target.value });
  };

  // Handle signup form changes
  const handleSignupChange = (e: ChangeEvent<HTMLInputElement>) => {
    setSignupData({ ...signupData, [e.target.name]: e.target.value });
  };
```

Figure 39 Login Code

```
const handleLoginSubmit = (e: FormEvent) => {
  e.preventDefault();
  if (loading) return;
  setLoading(true);

  login(loginData)
    .then((res: AxiosResponse) => {
      localStorage.setItem("token", res.data.token);
      localStorage.setItem("currentUser", JSON.stringify(res.data.userdata));
      navigate("/home");
    })
    .catch((error: AxiosError) => {
      const message = (error.response?.data as string) ?? "Server Error";
      toast.error(message);
    })
    .finally(() => setLoading(false));
};

return (
  <div className="aidflow-page">
    {/* Logo + Tagline */}
    <div className="aidflow-header">
      <h1 className="aidflow-logo">AidFlow</h1>
      <p className="aidflow-tagline">Empowering NGOs with Transparency.</p>
    </div>

    <div className="aidflow-container">
      {/* NGO Login Box */}
      <form class="aidflow-box" onSubmit={handleLoginSubmit}>
        <h2>NGO Login</h2>

        <input
          type="email"
          name="email"
          placeholder="Email"
          value={loginData.email}
          onChange={handleLoginChange}
          required
```

Figure 40 Login Code

```
return (
  <div className="aidflow-page">
    {/* Logo + Tagline */}
    <div className="aidflow-header">
      <h1 className="aidflow-logo">AidFlow</h1>
      <p className="aidflow-tagline">Empowering NGOs with Transparency.</p>
    </div>

    <div className="aidflow-container">
      {/* NGO Login Box */}
      <form className="aidflow-box" onSubmit={handleLoginSubmit}>
        <h2>NGO Login</h2>

        <input
          type="email"
          name="email"
          placeholder="Email"
          value={loginData.email}
          onChange={handleLoginChange}
          required
        />

        <input
          type="password"
          name="password"
          placeholder="Password"
          value={loginData.password}
          onChange={handleLoginChange}
          required
        />
```

Figure 41 Login Code

```
            <input
              type="text"
              name="registrationNumber"
              placeholder="Registration Number"
              value={signupData.registrationNumber}
              onChange={handleSignupChange}
              required
            />

            <input
              type="email"
              name="email"
              placeholder="Official Email"
              value={signupData.email}
              onChange={handleSignupChange}
              required
            />

            <input
              type="password"
              name="password"
              placeholder="Password"
              value={signupData.password}
              onChange={handleSignupChange}
              required
            />

            <button type="submit">Sign Up</button>
          </form>
        </div>
      </div>
    );
  }
```

Figure 42 Login Code

5.Analysis of progress

5.1 Progress Review

The first step in the development of AidFlow was the in-depth research of the donation environment in Nepal, the pain areas of the donors, and the transparency issues encountered by the NGOs. The problem description, literature study, requirement analysis, and feature breakdown were included in the interim documentation developed with the help of this foundation that also helped to formulate the system requirements. Moreover, much of the system design has been completed such as flowcharts, detailed wireframes, ERD, UML diagrams, and DFDs. These design artefacts were aligned with the user journeys of funders, NGOs and administrators to ensure smooth interactions between systems.

The development side has been configured with the MERN stack environment with the repository of the backend and frontend project. The database schemas using the ERD have been completed partially in MongoDB and the backend setup i.e., routing structure, environment variables, and base API have already begun. Most of the research, documentation and conceptual design work is already complete and the general progress rate is approximately 25 percent. Development of the backend has already begun, and the project is within the timeframe of the Gantt chart, which ensures that the following steps will be conducted on time.

5.2 Progress Table

Table 2 Progress Table

S.N.	Tasks	Status	Progress
1	Proposal Submission	Completed	100%
2	Interim Report	Completed	100%
3	Define Deliverables	Completed	100%
4	Create project Timeline	Completed	100%
5	Interim Report	Completed	100%
6	Gather NGO’s Data	Completed	100%

7	Gather requirements	Completed	100%
8	Analyse Transparency Needs	Completed	100%
9	Validate With Supervisor	Completed	100%
10	Finalize requirements	Completed	100%
11	UI/UX Wireframes	Completed	100%
12	Database Schema	Completed	100%
13	System Architecture	Completed	100%
14	Security Design	Completed	100%
15	UI Polishing + Testing (Design Phase)	Ongoing	20%
16	Sprint 1 – Authentication Module	Ongoing	10%
17	Sprint 2 – Donor Portal	Not Started	0%
18	Sprint 3 – NGO Portal	Not Started	0%
19	Sprint 4 – Admin Panel	Not Started	0%
20	Srint 5- Payment Gateway	Not Started	0%
21	Sprint 6 – Interacvtive map	Not Started	0%
22	Sprint 7 – Donation Tracking	Not Started	0%
23	Final Report Writing	Not Started	0%
24	Final Editing & Refinement	Not Started	0%
25	Deployment	Not Started	0%
26	Bug Fixes	Not Started	0%
27	Final Submission Preparation	Not Started	0%

5.3 Action Plan

Even though significant documentation and design tasks are done, there are various initial difficulties that influenced the speed of development. The last phase of executing the database structure-mostly verification flows, donation tracking, and update maintenance needed further analysis which brought a small delay when it came to back-end configuration. Not all UI and workflow decisions, including impact visualization and verification of the administration, did not need additional research, prolonging the design period they should have had in the planned sprint.

To solve these issues, the tasks that were left over in the previous sprints will be transferred to the next cycle. This involves completing all MongoDB schemas, the completion of the authorization and authentication, and the continuation of the development of the backend API. Moreover, the future effort will focus on frontend integration, initial testing, refinement of the UI screen, and system-wide integration preparation. The work can be done without disrupting the overall schedule of the project due to the iterative nature of Scrum which ensures that AidFlow moves in a consistent manner towards the next milestone of a functional prototype.

6 Future Work

6.1 Remaining Task Analysis with Gantt chart

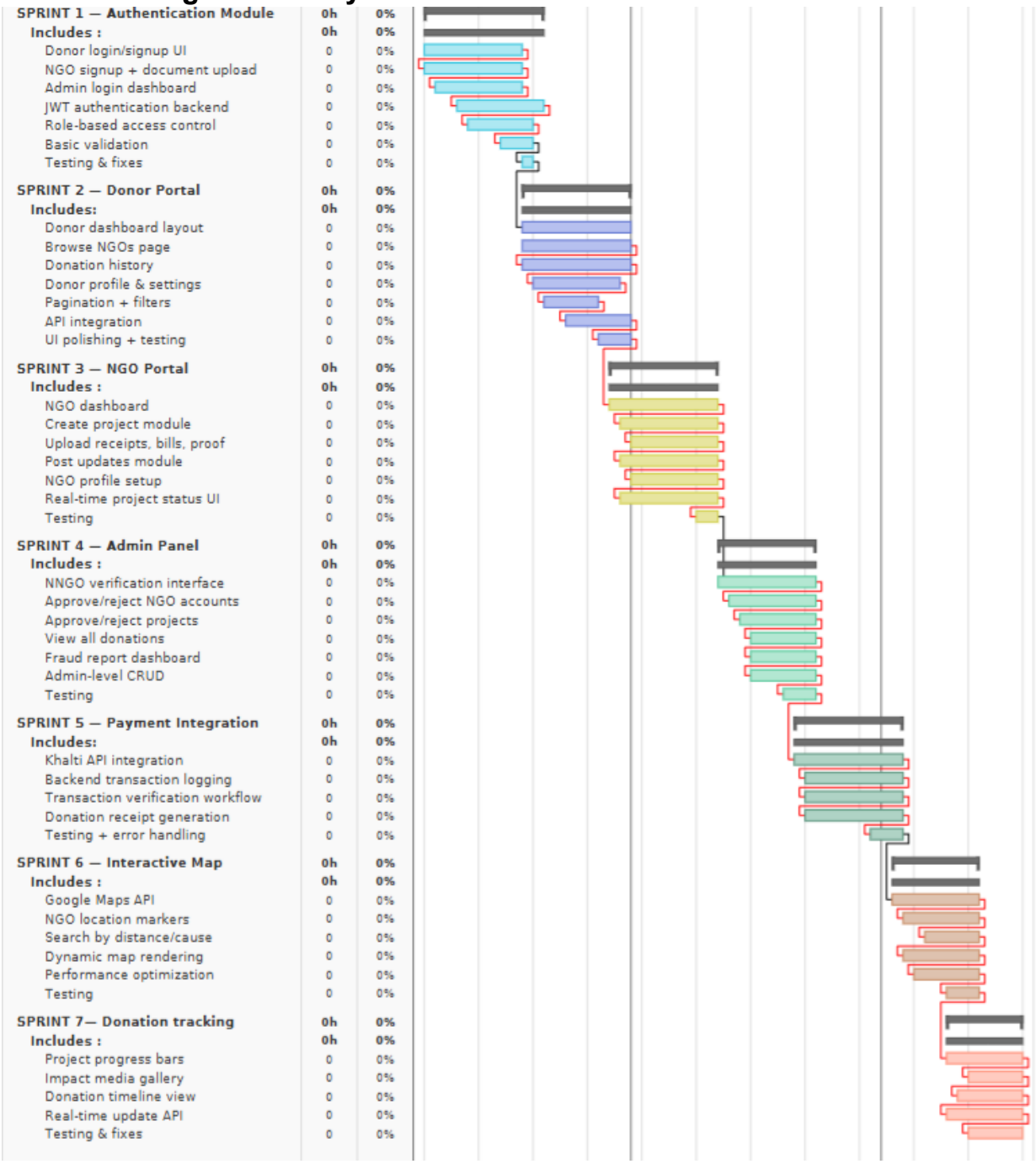

Figure 43 Remaining Work Gantt chart

6.1.1 For Further Future Work

Conclusion

The evolution of AidFlow has been a continuous process with good research, systematic records, and the comprehension of the issues in the donation system in Nepal. The interim phase was effective in laying the base of the system by analysing requirements, user journey, ERD design, UML diagrams, DFDs and initial architecture blueprint. All these factors together provide the guarantee that the fundamental goals of the platform transparency, trust, accountability, and accessibility are inculcated in the system.

The MERN stack environment is now configured, the backend framework is already established, and the necessary modules are planned to use the Agile sprints, which puts the project in the path of a smooth deployment in the next development cycles. The flexibility of Scrum coupled with action plan will make sure that unfinished tasks, e.g., NGO verification logic, donation tracking APIs, and dashboard construction are done in a systematic manner. When AidFlow reaches further points of its development, the emphasis will be on the incorporation of frontend elements, the optimization of workflows, and end-to-end testing. When it is ready, AidFlow should make a significant impact on the Nepalese philanthropic environment by creating a unique, transparent, and reliable platform that will allow closing the gap between donors and NGOs.

Bibliography

Bhandari, P. B. J. B. R., 2022. *Civil society and governance in Nepal: An analysis of NGO accountability.,* s.l.: s.n.

Boehm, 1988. *A spiral model of software development and enhancement.,* s.l.: s.n.

Charities Aid Foundation, 2023. *CAF World Giving Index,* s.l.: GALLUP.

Dingsøyr, T. N. S. B. V. a. M. N., 2012. *A decade of agile methodologies: Towards explaining agile software development.,* s.l.: s.n.

N. A. S., 2018. s.l.: s.n.

Prakash, A., 30 March 2010. *wiley.* [Online]
Available at: https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1748-5991.2009.01067.x

Shrestha, P. L. &. D. B., 2019. *A Nepalese Journal of ParϴcipatoryDevelopment,* kathmandu: Nepal Participatory Action Network (NEPAN).

Sommerville, 2016. s.l.: s.n.

Sommerville, I., 2016. *Software Engineering (10th ed.).,* s.l.: s.n.

S. W. C. A. R., 2022. s.l.: s.n.

Appendix

8.1 Appendix 1: System/Software Requirements (SRS)

8.1.1 Introduction

AidFlow is a web-based contribution management system that was developed to enhance charity giving in Nepal through accountability, openness, and trust. The platform can be regarded as transparent, allowing donors to be linked to vetted NGOs and community projects through real time updates, verified paperwork, visual proof of impact, and organisation of the donors and the recipients. AidFlow is aimed at resolving long-standing issues of lack of trust, visibility, and communications that have now become the hallmark of the donation landscape in Nepal.

8.1.1.1 Purpose

The objective of this SRS is to provide a comprehensive list of all the functional and non-functional requirements of the AidFlow platform. To the developers, the supervisor, evaluator, and stakeholders, this document will be the primary source of information to ensure that the features, limitations, and expected outcomes of the system are clearly defined. It outlines all system functionality necessary to have a reliable and efficient contribution management system such as donor, NGO, project, gift, verification, and transparency reporting.

8.1.1.2 Project Scope

This SRS is aimed at creating a general list of all functional and non-functional requirements to the AidFlow platform. This report is the main source of information to the developers, supervisors, evaluators, and stakeholders who will thoroughly understand the features of the system, constraints, and anticipated results. It describes all the characteristics required to construct a secure and effective contribution management system, including creating projects, processing gifts, registering donors, registering NGOs, verification procedures, and reporting transparency.

Major function:

- Registration and verification of NGOs.
- Registering donors and flow of donation.

- Creation, budgeting, updates, and uploading of proof-of-impact of projects.
- Donor real-time notifications and tracking.
- Approving and fraud control under administration.
- AidFlow will be transparent, user friendly, and will be able to foster trust in the philanthropic context of Nepal.

8.1.2 Overall Description

8.1.2.1 System Perspective Diagram

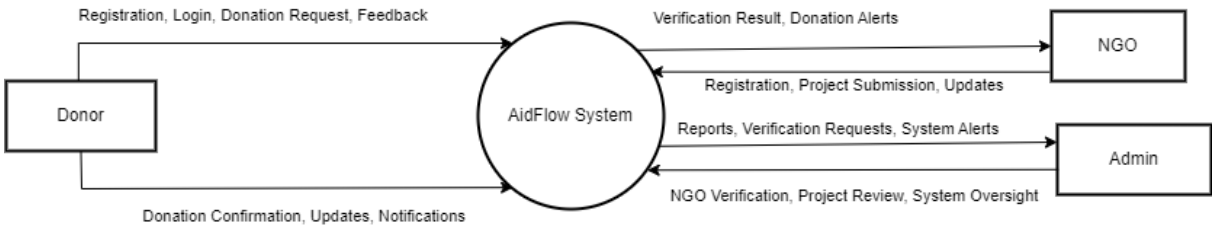

Figure 44 Overall Description

8.1.2.2 System Features

SF1: Registration (Donor & NGO)

- The personal details of the donors are self-registered.
- The registration involves NGOs filling their organizational documents to be verified.
- Before approval, verifying the NGO identity is done by Admin.

SF2: Login & Authentication

- Donor, NGO and admin secure logins.
- User authentication and session management
- Recover password

SF3: Project Management (NGO)

NGOs can:

- Develop new projects describing them with goals, budgets, and descriptions.
- Post receipts, bills, photos and reports.
- Report on project progress and status.

SF4: Donor Management (Donation)

Donors can:

- Browse NGOs and projects.
- Make donations safely through in-built payment platforms.

SF5: Evidence of Impact and Openness.

NGOs post:

- Bills
- Invoices
- Before and after pictures
- Updates in the field
- Every donor will be able to observe these changes.

SF6: Administrative Checking System.

- Administrators can:
- Accept or reject NGOs
- Accept or reject projects

SF7: Alerts

- The system transmits:
- Confirmations of donations
- Alerts for project updates
- Verification outcomes
- Results of NGO approval

SF8: Impact Dashboard

- Donors can:
- Keep track of every donation
- See the status of project**.**

8.1.2.3 User Classes and Characteristics

8.1.3 Operating Environment

- **IDE:** Visual Studio Code
- **Frontend Framework:** React.js
- **Backend Runtime Environment:** Node.js
- **Backend Framework:** Express.js
- **Database:** MongoDB
- **Version Control:** GitHub
- **API Testing Tool:** Postman

8.1.4 Design and Implementation

- It should be based on MERN architecture.
- Clear audit records should be maintained.
- Stored files should include receipts and photos.
- The NGOs will have to be verified before the publication of their projects.
- The payment gateway should be compatible.
- Need to be able to handle large volumes of image upload.
- Should also comply with data protection and privacy laws.

8.1.5 Assumptions and Dependencies

Assumptions

- The internet is accessible to NGOs and donors.
- NGOs can offer legitimate registration documents.
- Customers provide valid personal information.
- Payment gateways are reliable.
- The users have skeleton digital literacy.

Dependencies

- Availability of the MongoDB databases.
- Notification and OTP email service.
- Payment gateways integration.
- Browsers Compatibility of modern JS frameworks.

8.1.3 Functional Requirements

Registration & Login

Table 3 Registration & Login

Req ID	Requirement	Priority	Complexity
RG01	System should permit the registration of donors	Must	Low
RG02	System should enable NGO registration through uploading of documents	Must	High
RG03	Admin should check the profiles of NGOs	Must	Medium
RG04	Secure log-in of the users is required.	Must	Medium
RG05	System should help with resetting passwords	Should	Medium

Project and Donation Management

Table 4 Project and Donation

Req ID	Requirement	Priority	Complexity
PM01	Should form projects by NGOs	Must	Medium
PM02	NGOs should update with media uploads	Must	High
PM03	Projects donation required of donors	Must	Medium
PM04	Generation of donation receipt	Must	Medium
PM05	Donation history can be viewed by donors	Must	Low

Verification System

Table 5 Verification System

Req ID	Requirement	Priority	Complexity
VS01	Approve/ reject NGOs in the permission of the admin	Must	Medium
VS02	Project documentation has to be checked by the administrator	Must	Medium
VS03	System should alert of suspicious documents	Should	High

Transparency & Impact Reporting

Table 6 Transparency & Impact Reporting

Req ID	Requirement	Priority	Complexity
VS01	NGO must upload the proof of impact	Must	High
VS02	Donors must view project progress	Must	Medium
VS03	Donors can download receipts and reports	Should	Medium

8.1.4 External Interfaces

8.1.4.1 User Interface

Responsive React interface

- Role-based UI Donor, NGO, Admin.
- Dashboard project cards charts, charts.
- Clean project progress colour coded.

8.1.4.2 Hardware Interfaces

- Keyboard, mouse, touchscreen

- NGOs Camera/file uploader.

8.1.4.3 Software Interfaces

- Payment gateway (Khalti / Stripe)
- Email SMTP
- MongoDB Atlas
- Cloud storage for media

8.1.4.4 Communication Interfaces

- HTTPS encryption
- REST API

8.1.4.5 Non-Functional requirements

Non-Functional requirements are the requirements which are not tied to functions.

Performance

- Load dashboard in < 5 seconds
- Handle 5000+ users concurrently

Security

- JWT authentication
- Encrypted password storage
- Role-based access control

Reliability

- Daily data backup of databases.
- 24/7 availability

8.1.4.6 Safety Requirements

- Sensitive data are encrypted when they are being transmitted and stored.
- 30-minute session time out.
- None of the malicious scripts and malware.

8.3 Appendix 3: High Level and Expanded Level Descriptions

High Level Use Case

The system as seen by the end user is the top level use cases that are description of the key characteristics. Therse visuals and explanations provide a simplified overview of the way the system of AidFlow works getting a simplified overview of the way the system of AidFlow worked without getting into any technical details. They attract attention to the major functions of the platform, participants, and the overall workflow. All the high-level use cases represent the main exchanges that occur within the system between donors, NGOs and administrators.

1. **Use Case**: Access the System

   Actors: Donor, NGO, Admin

   Description: This use case describes the way positive credentials are used to access the AidFlow system. The system permits access to the user based on his or her responsibilities after verifying the identity of the user.
2. **Use Case**: register to the system

   Actors: Admin (verifier), NGO (initiator)and Donor(initiator).

   Description: To open an account, unregistered users (nongovernmental organization and donors) are required to provide the required information. The administrator grants or rejects registration of NGOs after going through the paperwork provided.)
3. **Use Case:** Browse NGOs and Projects Use Case.

   Actors: Donor

   Description: This use allows donors to visit and browse through variable NGOs and view their activities listed with description, goals, and information on their impact.
4. **Use Case:** Contribution to the cause

   Actors: Donor(initiator)

   Description: This use case provides donors an opportunity to safely donate to any listed AidFlow initiative. The system generates receipts and pays.
5. **Use Case:** Project management

Actor: NGO

Description: NGOs can administer the project status (active, completed, validated), post proof papers, create new projects, and update project information.

6. **Use Case:** Uploading Impact

   Actor: NGO

   Description: NGOs upload records, photographs, reports on progress, and their impact. To be transparent, funders are presented with these changes.

7. **Use Case:** Verification and approval

   Actor: Admin

   Description: The responsibilities of the administrator are to verify the uploaded papers, analyse the submitters of the projects, approve or disapprove the requests and verify the accounts of the NGOs.

8. **Use Case:** Track Donations

   Actor: Admin, Donor, NGO

   Description: Donors can monitor their donation records and project. The received donations are monitored by NGOs, and the admins keep track of donations logs to ensure transparency.

9. **Use Case:** System Notification

   Actor: Admin, Donor, NGO

   Description: The system sends automatic messages on updates, verifications, and donation confirmations as well as changes in project status.

Expanded Use Case

Use Case: Browse NGOs and Projects Use Case.

Actors: Donor

Description: This use allows donors to visit and browse through variable NGOs and view their activities listed with description, goals, and information on their impact.

Table 7 Expanded Use case

Actor Action	System Response
Donor selects AidFlow and clicks on either Browse NGOs or Browse Projects.	The system gives all the verified NGOs/projects and shows them.
Donor filters by category, location or impact area.	System filters and updates project listing.
Donor chooses a project to see information.	The system shows the project description, budget, updates and verification status.
Donor selects the options of Donate and fills in payment details.	System authenticates information and makes the payment.
Donation is registered in database	System e-mails a donation confirmation and receipt to the donor.

Alternative Course

At Step 3:

In case no projects are found in the filter, the system displays "No projects found."

At Step 7:

The donor can always terminate the process if he or she terminates payment.

Use Case: Uploading Impact

Actors: NGO (initiator)

Description: Through this use case, NGOs can post impact reports, receipts, photos and updates to demonstrate how donor money is being spent.

Table 8 Expanded Use case

Actor Action	System Response
NGO logs into the system and goes to the project updates	The update upload form is shown in the system.

NGO posts images, documentations, descriptions, or reports of progress.	The system authenticates metadata and uploaded files.
NGO submits the update.	The update is saved in the database by the system.
System informs the donors after the project.	The NGO is informed through system confirmation.

Alternative Courses

At Step 3:

In case the NGO posts half of the information, the update is declined.

At Step 6:

On failure of upload, system warns NGO and records failure.

8.4 Appendix 4: Future Work

8.4.1 Remaining Phases

Backend Sprints

- NGO verification logic
- Generation and revision modules of projects.
- Donation tracking APIs
- Transparency reporting

Frontend Sprints

- Frontend Sprints

Donor dashboard

NGO dashboard

Admin approval panel

Real-time project timeline

8.4.2 Detailed Plan

The rest of the development will be built on Agile Sprints:

- Complete back-end authentication, donation API.
- Installation project manager and update modules.
- Get front and back on the same page.
- Test donor - NGO - workflow of administration.
- Deploy frontend + backend
- Final testing & submission

8.5 Appendix 5: Survey Form

Survey Form Screenshot

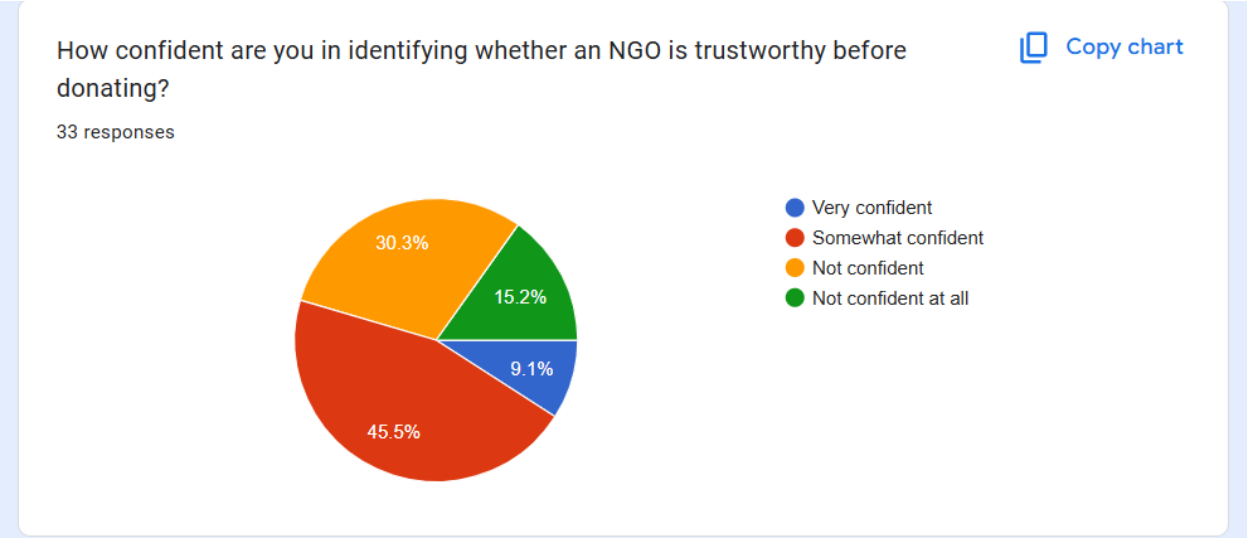

Figure 45 Survey Form

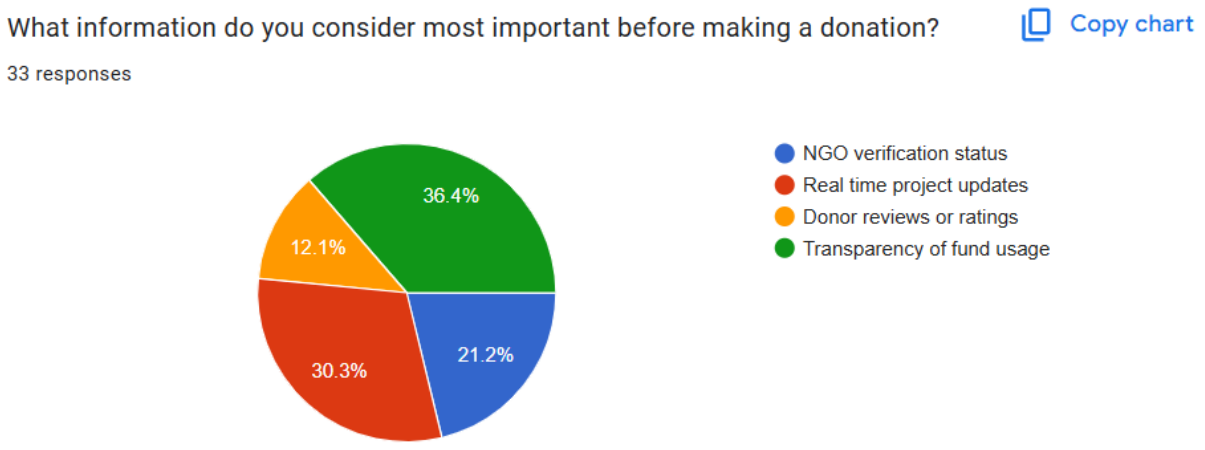

Figure 46 Survey Form

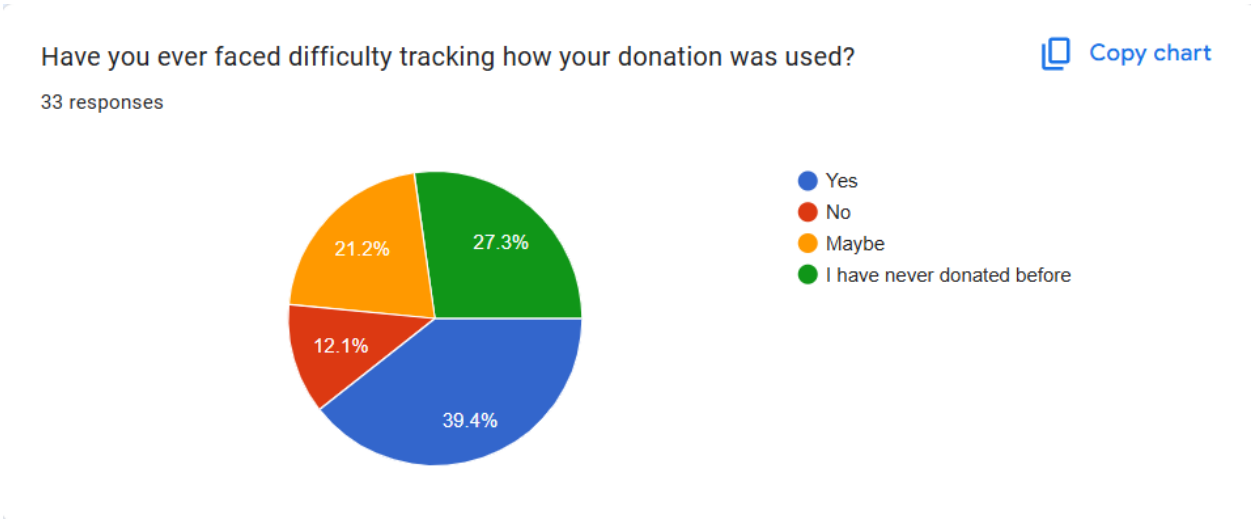

Figure 47 Survey Form

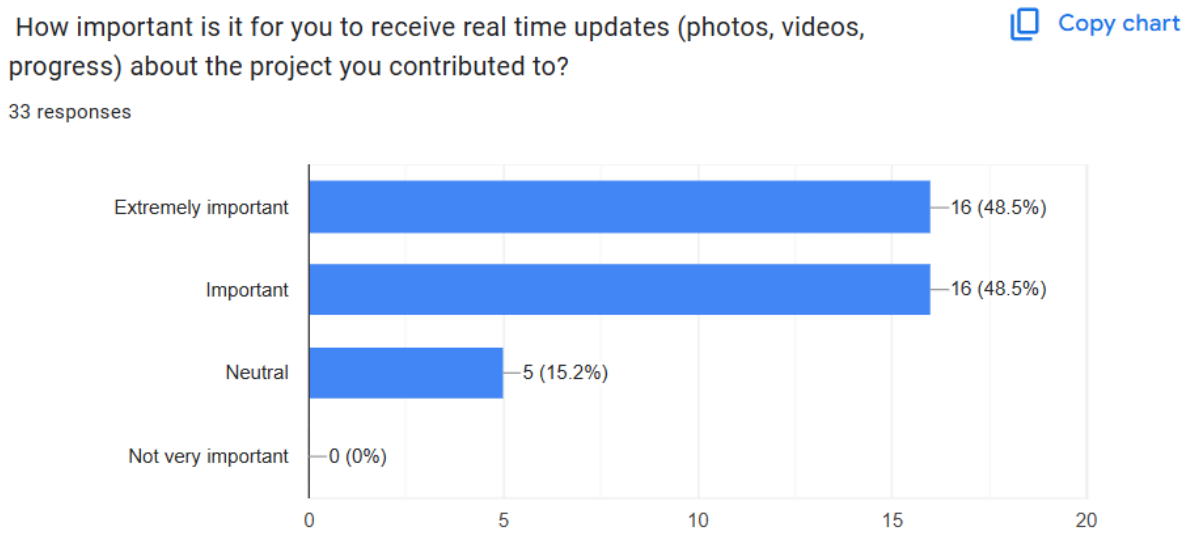

Figure 48 Survey Form

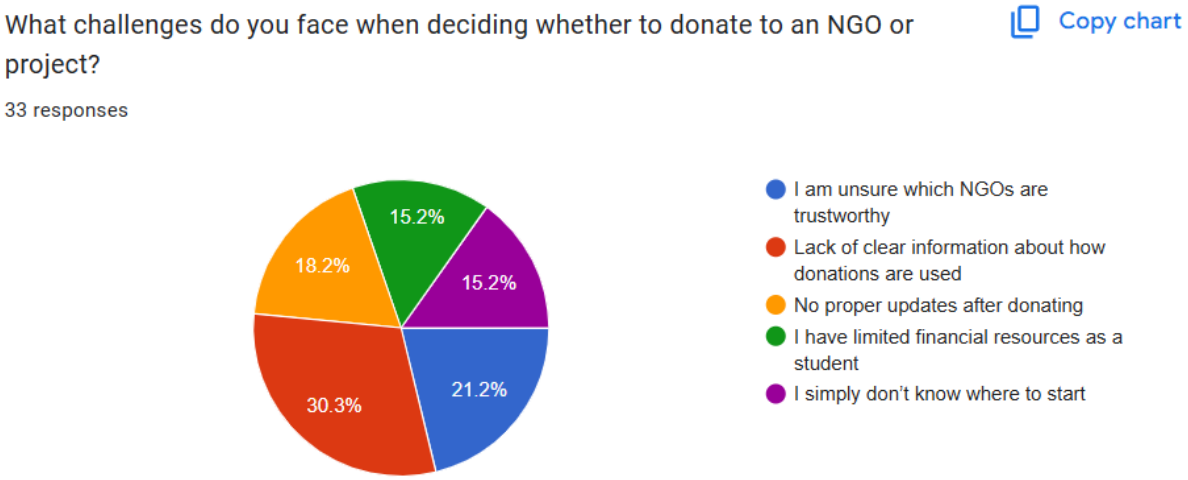

Figure 49 Survey Form

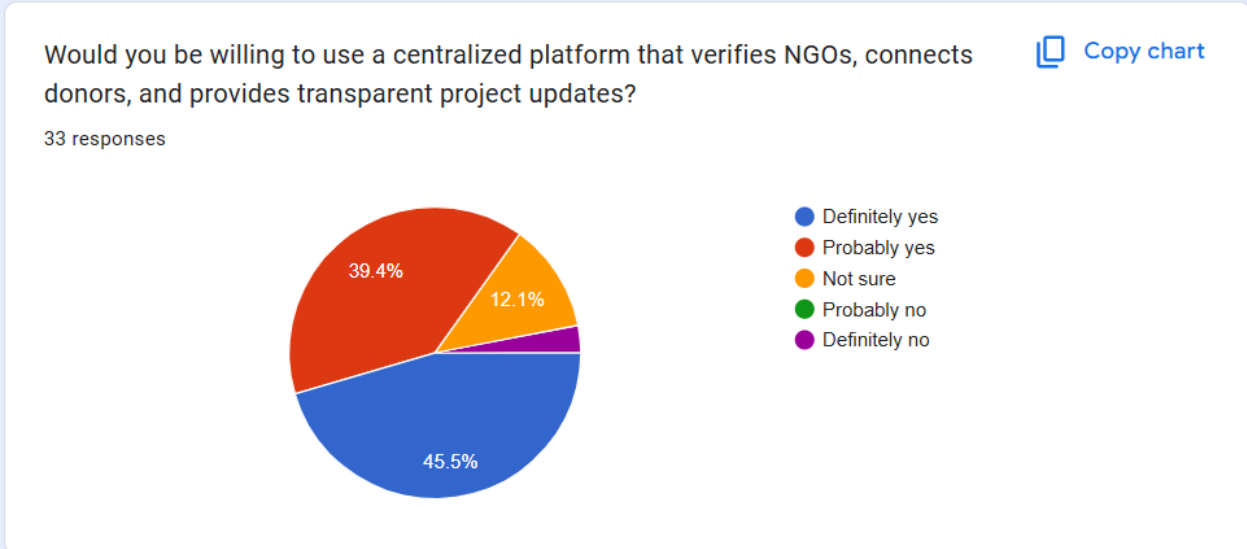

Figure 50 Survey Form

8.6 Appendix 6: Used Methodology

Agile Methodology (Scrum Framework)

Agile Scrum approach has been selected because it promotes flexibility, user feedback and incremental development which is most appropriate in a highly dynamic system like in a DMS. Agile creates room to change plans, continuous improvement, and by bypassing this, one developer will do a good job in keeping up with the project changes (Dingsøyr, 2012).

Scrum divides the work into a series of brief sprints, the results of which are operating elements, e.g. payment integration, donor verification, NGO onboarding or real-time updates.

Benefits of Scrum

Incremental: Scrum helps in the delivery of working modules after each sprint which enhances the eventual results in terms of quality by refinement.

High Flexibility: Agile permits implementation of new requirements into the development flow and it does not interrupt the development process.

User Centred Approach: A review is undertaken on a regular basis to make sure that the donor, NGO and supervisor responses are included in all versions.

Enhanced Productivity: The pressure in the workload is minimized because the workload is broken into manageable segments and the gradual improvement is achieved.

Less Risk: The frequentness of the testing in each sprint is employed to identify the problems at an early stage and avoid significant failures in the creation process of the development process.3.3 Work Breakdown Structure (WBS)

8.7 Appendix 7: Problem Statement

Supporting Data

- Transparency fosters trust of donors.

  It has been shown by the international studies that an international donor is much better likely to make the donation when the charities show valid impact report and financial transparency. Transparent reporting and transparency are directly associated with the long-term giving and the guarantee of the donors (Charities Aid Foundation, 2023).
- The credibility is compromised concerning the information misrepresentation to the donor.

  The reason why many people among the population are not willing to donate is because they lack the understanding of how their money is being squandered by the NGOs (governance studies carried out in Nepal).This lack of clarity is something that discourages their participation and acts as a barrier to the philanthropic contribution. (NEPAL ADMINISTRATIVE STAFFCOLLEGE, 2018).
- Lack of proper communication divides the inputs and the outputs.

  Development organisations reports and in the context of reports, there are indications that ineffective reporting systems and intermittent nature of

news that NGOs feed on leave donors in a sense of disconnection with the real effects. (Bhandari, 2022).

- Nepal NGO sector is obscure regardless of the size of the sector.

  Tens of thousands of registered Non-Governmental Organizations in Nepal have minimal or no transparency, accountability or disclosure. The donors are faced with a problem of locating legitimate organizations. (Social Welfare Council Annual Report, 2022).